Development of a Double-b Tagger for Boosted Higgs Jets

A dissertation submitted for the partial fulfilment of BS-MS dual degree in Physics

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Declaration

The work presented in this dissertation has been carried out by me under the guidance of **Dr. Jyothsna Rani Komaragiri** at the **Indian Institute of Science (IISc), Bengaluru**.

This work has not been submitted in part or in full for a degree, a diploma, or a fellowship to any other university or institute. Whenever contributions of others are involved, every effort is made to indicate this clearly, with due acknowledgement of collaborative research and discussions. This thesis is a bonafide record of original work done by me and all sources listed within have been detailed in the bibliography.



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CERTIFICATE

This is to certify that Abhiraaj Baishya, student of BS-MS (10th Semester) in Physical Sciences of Indian Institute of Science Education and Research, Berhampur has carried out the dissertation titled, "Development of a Double-b Tagger for Boosted Higgs Jets" under my supervision for the partial fulfilment of his BS-MS Degree in Physical Sciences of this Institute.

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Abstract

The identification of Higgs boson decaying into a pair of bottom quarks ($H \rightarrow b\bar{b}$) at boosted regimes is one of the key tasks facing both the CMS and ATLAS experiments at the LHC collider. In the case of boosted regime, the decay products combine to form one wide jet, making the QCD background suppression problem highly challenging.

In order to go beyond these constraints, we have developed a highly advanced Multi-Input Deep Neural Network (DNN). The network consisted of four data channels: global jet information, particle flow (PF) candidates, tracks, and secondary vertices. The use of 1D-Convolutional layers allowed us to perform the feature extraction process directly from the low-level constituents, which made the DNN reach an impressive AUC score of 0.98. In addition, the analysis of feature importance using a variety of permutation approaches showed that using the particle-level information available in the PF channel gives the highest gain in the sensitivity of the model. Thus, our studies have proved the usefulness of the developed tagger as an instrument that can effectively improve the sensitivity of the $H \rightarrow b\bar{b}$ search in the boosted regime.

Keywords: Boosted Higgs, Jet Substructure, Double-b Tagging, CMS Open Data, BDT, Multi-Input DNN.

Contents

1 Introduction **1**

1.1 The Yukawa Coupling and $H \to b\bar{b}$ 1

1.2 The Boosted Regime and FatJets 1

1.3 Motivation: From BDT to Deep Learning 2

1.4 Thesis Objectives and Scope 3

2 Theoretical Framework **4**

2.1 The Higgs Sector and Yukawa Couplings 4

2.1.1 The Yukawa Lagrangian 4

2.1.2 Branching Fractions and the Dominance of $H \to b\bar{b}$ 5

2.2 Quantum Chromodynamics and Jet Physics 5

2.2.1 The Anti-k_T Clustering Algorithm 5

2.2.2 Jet Grooming: The SoftDrop Technique 8

2.3 Physics of B-Hadron Identification 9

2.3.1 Secondary Vertices (SV) 9

2.3.2 Impact Parameter (IP) Significance 10

2.3.3 N-Subjettiness (τ_N) 10

3 Dataset and Feature Analysis **12**

3.1 Simulated Samples and Event Selection 12

3.2 The Global FatJet Branch (38 Variables) 12

3.2.1 Mass Reconstruction and SoftDrop Grooming 13

3.2.2 Jet Substructure: N-Subjettiness 13

3.3 The Particle Flow (PF) Candidate Branch (24 Variables) 13
3.3.1 Kinematic Variables: $p_{T,rel}$ and ΔR 13
3.3.2 Displacement Signatures 14
3.4 The Secondary Vertex (SV) Branch (11 Variables) 14
3.4.1 SV Mass and Flight Distance 14
3.4.2 Vertex Geometry: $\cos(\theta)$ 14
3.5 The Track Branch (6 Variables) 14
3.5.1 Impact Parameter Significance 15
3.5.2 Track Kinematics and Quality 15
3.6 Detailed Feature Definitions 15
3.6.1 Branch 1: Global FatJet Features (38 Variables) 15
3.6.2 Branch 2: Particle Flow (PF) Candidate Features (24 Variables) . . 16
3.6.3 Branch 3: Track-Level Features (30 Variables) 17
3.6.4 Branch 4: Secondary Vertex (SV) Features (11 Variables) 18
3.7 Analysis of Feature Sensitivity 19
3.7.1 Synergy Between Tracks and Vertices 19
3.7.2 Training Labels and Truth Definitions 19
3.7.3 Phase Space Selection and Spectator Variables 20
3.7.4 Implementation of Filtering Logic 21

4 Methodology **22**
4.1 Phase I: Gradient Boosted Decision Trees Architecture 22
4.1.1 The Rationale for Gradient Boosting with XGBoost 22
4.1.2 Data Pipeline and Feature Engineering 23
4.1.3 Feature Extraction with Uproot 23
4.1.4 Data Formatting for XGBoost 23
4.1.5 Model Architecture and Hyperparameter Optimization 23
4.2 Phase I: BDT Performance Evaluation Methodology 25
4.2.1 Discriminant Score Distribution 25
4.2.2 Threshold-Independent Metrics: ROC and AUC 25
4.2.3 Kinematic Robustness and Binning 25
4.2.4 Operating Point Analysis: The Confusion Matrix 26
4.2.5 Motivation for Phase II 26
4.3 Phase II: Multi-Input Deep Learning Architecture 26
4.3.1 Data Acquisition and Pre-processing Pipeline 26
4.3.2 Standardizing Raw ROOT Data 26
4.3.3 Feature Engineering and Selection Logic 27
4.3.4 Working with PyTorch DataLoaders 27
4.3.5 Multi-Stream Model Architecture 27

4.3.6 Batch Construction and Data Collation 28
4.3.7 Feature Fusion and Classification Head 28
4.3.8 Optimization and Binary Logic 29
4.4 Phase II: Method of Evaluation for DNNs Performance 29
4.4.1 Diagnostic Monitoring of Training Dynamics 30
4.4.2 Discriminant Analysis and Polarization 30
4.4.3 Statistical Benchmarking via ROC Analysis 30

5 Results and Performance Analysis **31**
5.1 Phase I: BDT Baseline Evaluation 31
5.1.1 Standalone BDT ROC Analysis 31
5.1.2 Comparison with Default CMS Tagger 32
5.1.3 Performance Stability in SoftDrop Mass Bins 33
5.1.4 Performance Dependence on Transverse Momentum (p_T) 34
5.1.5 Classification Accuracy and Error Analysis 35
5.1.6 Classifier Discriminant Analysis 36
5.1.7 Interpretability and Feature Importance 37
5.2 Limitations of the Phase I BDT Baseline 38
5.2.1 The ”Fixed-Feature” Information Loss 39
5.2.2 The High-p_T Resolution Challenge 39
5.2.3 Feature Interaction Complexity 39
5.3 Phase II: Multi-Input Deep Neural Network (DNN) 39
5.3.1 Overall DNN Performance and ROC Analysis 40
5.3.2 Benchmarking: DNN vs. Default Double-B Tagger 41
5.3.3 Kinematic Stability and Binning Analysis 42
5.3.4 Integrated Sensitivity and Feature Interpretability 43
5.3.5 Detailed Classification Accuracy: Raw Counts vs. Percentages . . . 45
5.3.6 Discriminant Distributions and Classification Power 46
5.4 Comparative Performance Analysis 48
5.5 Final discussion of Results 48

6 Conclusion **50**
6.1 Summary of Findings 50
6.2 Technical Constraints and Architectural Choices 51
6.3 Final Reflections 51

Bibliography **51**

List of Figures

1.1 A pictorial representation of jets (green cones) arising from a W, Z, or Higgs boson, decaying to two quarks with increasing Lorentz boost. For highly boosted bosons the two quarks are emitted very close to each other, giving rise to one single, wide jet with substructures (blue cones) that correspond to the two original quarks. Image taken from https://cms.cern/news/w-z-and-higgs-bosons-portals-exotic-physics 2

4.1 Code snippet showing my BDT model architecture 24
4.2 Code Snippet of my DNN model architecture 29

5.1 Standalone ROC curve for the Phase I BDT. The Area Under the Curve (AUC) of 0.971 indicates strong baseline discrimination. In practice, this level of performance already allows for a meaningful separation between signal and background, although there is still visible overlap in the intermediate score region.. 32
5.2 Performance comparison between the custom BDT (red) and the standard CMS DeepDoubleB algorithm (blue). The strong agreement validates the feature selection and training parameters used in this study. 33
5.3 BDT ROC curves categorized by m_{SD} bins. The stability of the curves across different mass ranges indicates that the BDT is learning flavor and substructure features rather than just the mass resonance. 34

5.4 BDT ROC performance across different transverse momentum bins. A slight degradation in background rejection is observed at extremely high p_T, where track separation reaches the limit of the silicon tracker's resolution. 35
5.5 Confusion matrices for the Phase I BDT evaluated on the independent test set. These metrics quantify the model's ability to isolate the $H \to b\bar{b}$ signal from the inclusive QCD background. 36
5.6 Normalized BDT output score (Discriminant) for $H \to b\bar{b}$ signal (red) and QCD background (blue). The logarithmic scale reveals the efficiency of the model in the extreme signal-like region (> 0.9). 37
5.7 Feature importance ranking for the top 20 variables in the Phase I BDT. The importance score reflects the total gain contributed by each feature across all decision trees. 38
5.8 Overall Receiver Operating Characteristic (ROC) curve for the Phase II Multi-Input DNN. The model achieves an integrated AUC of 0.9818 at $\sqrt{s} = 13$ TeV. 40
5.9 Performance comparison between the proposed Multi-Input DNN (AUC = 0.982) and the standard CMS baseline (AUC = 0.916). 41
5.10 DNN performance binned by p_T. The model shows remarkable stability, with the $350 < p_T < 600$ GeV bin achieving a peak AUC of 0.984. 42
5.11 DNN performance binned by SoftDrop mass (m_{SD}). The highest performance (AUC = 0.978) is observed in the $120 < m_{SD} < 160$ GeV bin, coinciding with the Higgs mass window. 43
5.12 Global branch-wise sensitivity. 44
5.13 Top 25 integrated features. 44
5.14 feature importance in different input branches. 44
5.15 Absolute Confusion Matrix (Raw Counts) 46
5.16 Normalized Confusion Matrix (%) 46
5.17 Comparative view of the Phase II DNN confusion matrices. This side-by-side layout provides a complete overview of the model's performance on the large-scale test dataset. 46
5.18 Distribution of the DNN output scores for the signal ($H \to b\bar{b}$) and background (QCD) samples. The clear separation between the two peaks at the boundaries of the unit interval demonstrates the high classification power of the multi-stream architecture. 47
5.19 Comparison of ROC performance for the Phase II DNN, Phase I BDT, and the CMS Double-B baseline. The log-scale plot (right) highlights the superior background rejection in the high-purity regime. 48

List of Tables

3.1 Global Jet-Level Features and Physical Significance 16
3.2 PF Candidate Features (Per-Constituent) 17
3.3 Detailed Track-Level Features for Boosted b-Tagging 17
3.4 Secondary Vertexing Features and Geometric Observables 18
3.5 Binary Training Labels and Logic 20
3.6 Kinematic Constraints and Spectator Variables 20

5.1 Comparison of signal efficiencies (%) for $H \to b\bar{b}$ tagging at fixed background efficiency benchmarks. 48

List of Abbreviations

AUC	Area Under the Curve
BDT	Boosted Decision Tree
BSM	Beyond the Standard Model
CMS	Compact Muon Solenoid
CNN	Convolutional Neural Network
DNN	Deep Neural Network
HL-LHC	High-Luminosity Large Hadron Collider
IP	Impact Parameter
LHC	Large Hadron Collider
MC	Monte Carlo
ML	Machine Learning
PF	Particle Flow
QCD	Quantum Chromodynamics
ReLU	Rectified Linear Unit
ROC	Receiver Operating Characteristic
SM	Standard Model
SV	Secondary Vertex

Chapter 1

Introduction

The validation of the Standard Model (SM) of particle physics reached a definitive milestone in 2012 with the discovery of the Higgs boson by the ATLAS and CMS collaborations at the Large Hadron Collider (LHC). This scalar particle, associated with the Brout-Englert-Higgs mechanism, provides a fundamental explanation for the origin of mass in elementary particles via electroweak symmetry breaking. Since this discovery, the primary objective of the high-energy physics community has transitioned from observation to high-precision spectroscopy of the Higgs sector.

1.1 The Yukawa Coupling and $H \to b\bar{b}$

A central component of this precision era is the measurement of the Higgs boson's Yukawa couplings to fermions. Among these, the decay of the Higgs boson into a pair of bottom quarks ($H \to b\bar{b}$) is of paramount importance. With a branching fraction of approximately 58% for a Higgs mass of 125 GeV, it is the dominant decay mode. However, despite its high rate, the $H \to b\bar{b}$ channel is experimentally obscured by an overwhelming background from Quantum Chromodynamic (QCD) multijet production, which possesses a cross-section several orders of magnitude higher.

1.2 The Boosted Regime and FatJets

As the LHC reaches higher center-of-mass energies and luminosities, the search for New Physics (BSM) and rare SM processes often points toward the "boosted" regime. In this

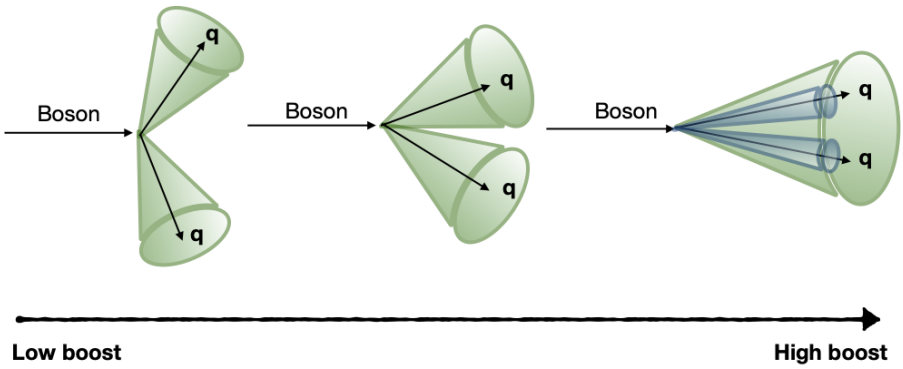


Figure 1.1: A pictorial representation of jets (green cones) arising from a W, Z, or Higgs boson, decaying to two quarks with increasing Lorentz boost. For highly boosted bosons the two quarks are emitted very close to each other, giving rise to one single, wide jet with substructures (blue cones) that correspond to the two original quarks. Image taken from https://cms.cern/news/w-z-and-higgs-bosons-portals-exotic-physics

scenario, the Higgs boson is produced with a high transverse momentum ($p_T > 250$ GeV), typically recoiling against a high-energy vector boson or a jet.

Due to the Lorentz boost, the angular separation ΔR between the two bottom quarks is roughly proportional to $2m_H/p_T$. At these high momenta, the decay products become highly collimated, merging into a single large-radius jet, or "FatJet." Traditional b-tagging algorithms, which were designed to resolve individual jets from isolated quarks, suffer from a significant loss in efficiency in this merged topology. The challenge, therefore, lies in developing "Double-b" tagging algorithms capable of looking inside the FatJet to identify the two-pronged substructure and the flavor of the constituents.

1.3 Motivation: From BDT to Deep Learning

In the preliminary phase of this research (the Mid-Thesis report), a baseline was established using a Boosted Decision Tree (BDT). This model utilized 38 high-level "expert" variables—mathematical summaries of the jet's shape and energy distribution. While the BDT served as an effective initial discriminator, it inherently assumed that these 38 variables contain all the necessary information for the required classification.

However, this assumption creates an information bottleneck. Raw detector data, containing hundreds of individual Particle Flow (PF) candidates, tracks, and secondary vertices, has complex non-linear correlations that high-level summaries cannot fully capture. This thesis shows the transition to a Multi-Input Deep Neural Network (DNN) architecture. By taking the help of 1D-Convolutional layers and multiple branches that consist of different data streams, the proposed DNN aims to achieve maximum sensitivity from the low-level constituents of the jet.

1.4 Thesis Objectives and Scope

This work aims to build, train, and validate a high performing Deep Learning tagger for boosted Higgs identification. The scope includes:

- Extraction and initial processing of CMS-simulated datasets for $H \rightarrow b\bar{b}$ signal and QCD background.
- Comparative analysis of the Phase I BDT with the Phase II Multi-Input DNN .
- Analysis of feature importance to interpret the mysterious nature of the neural network and validate its learning with established particle physics principles.
- Evaluation of performance of the model through standard metrics, including ROC curves and Area Under the Curve (AUC).

Chapter 2

Theoretical Framework

Before getting to the tagging methodologies, it is important to be aware of the physical and mathematical foundation necessary to understand the $H \rightarrow b\bar{b}$ decay process and the reconstruction of jet substructure at the Large Hadron Collider (LHC). This chapter is aimed at serving this purpose.

2.1 The Higgs Sector and Yukawa Couplings

In the Standard Model (SM), the Higgs boson (H) is the physical scalar manifestation of the Brout-Englert-Higgs field. This field is unique in the SM as it possesses a non-zero **Vacuum Expectation Value (VEV)**, denoted as $v \approx 246$ GeV. Through the process of spontaneous symmetry breaking, the Higgs field provides mass to the $W^{\pm}$ and Z gauge bosons, while its interaction with fermions (quarks and leptons) generates their masses via Yukawa couplings.

2.1.1 The Yukawa Lagrangian

The interaction between the Higgs field and the fermion fields ψ_f is described by the Yukawa Lagrangian. In the unitary gauge, after the symmetry breaking, the Lagrangian term for a fermion f can be expressed as:

$$\mathcal{L}_{Yukawa} = -\sum_f \frac{m_f}{v} \bar{\psi}_f \psi_f H = -\sum_f g_{ffH} \bar{\psi}_f \psi_f H \tag{2.1}$$

where:

- m_f is the mass of the fermion.
- v is the vacuum expectation value of the Higgs field.
- $g_{ffH} = m_f/v$ is the **Yukawa coupling constant**.

A fundamental prediction of the Standard Model is that the coupling strength g_{ffH} is directly proportional to the mass of the fermion. This leads to a distinct hierarchy: the Higgs boson couples most strongly to the heaviest particles.

2.1.2 Branching Fractions and the Dominance of $H \to b\bar{b}$

The decay width of the Higgs boson into a fermion-antifermion pair ($H \to f\bar{f}$) is proportional to the square of the Yukawa coupling (g^2_{ffH}). For quarks, accounting for the three color degrees of freedom and QCD radiative corrections, the partial width is given by:

$$\Gamma(H \to q\bar{q}) = \frac{3G_F m_H m_q^2(m_H)}{4\pi\sqrt{2}} (1 + \Delta_{QCD}) \tag{2.2}$$

Because the top quark ($m_t \approx 173$ GeV) is heavier than half the Higgs mass ($m_H/2 \approx 62.5$ GeV), the decay $H \to t\bar{t}$ is kinematically forbidden for a on-shell Higgs boson. Consequently, the Higgs boson predominantly decays into the next heaviest accessible fermion: the **bottom quark** (b), which has a running mass of $m_b(m_H) \approx 2.8$ GeV (pole mass ≈ 4.18 GeV).

For a Higgs mass of 125 GeV, the $H \to b\bar{b}$ branching fraction is approximately **58.1%**. Despite being the most frequent decay mode, it remained one of the most challenging to observe due to the immense background from multijet QCD processes. Identifying this decay, particularly in the boosted regime where the two b-quarks merge into a single jet, is the primary motivation for the advanced tagging techniques developed in this thesis.

2.2 Quantum Chromodynamics and Jet Physics

Quarks produced in high-energy collisions cannot exist as free particles due to "color confinement." Instead, they undergo a process of fragmentation and hadronization, creating a collimated spray of particles known as a "jet."

2.2.1 The Anti-k_T Clustering Algorithm

To study these jets, we use sequential recombination algorithms. The CMS experiment primarily utilizes the **Anti-k_T algorithm**, which is defined by its ability to produce geometrically regular, "circular" jets that are resilient to the effects of the underlying event and pileup.

Mathematical Definition and Distance Measures

The algorithm relies on two specific distance metrics to determine the order in which particles are clustered. For every pair of particles (or clusters) i and j, the algorithm calculates the iterative distance d_{ij} and the beam distance d_{iB} as follows:

$$d_{ij} = \min(k_{ti}^{-2}, k_{tj}^{-2}) \frac{\Delta R_{ij}^2}{R^2} \tag{2.3}$$

$$d_{iB} = k_{ti}^{-2} \tag{2.4}$$

Where the terms are defined as:

- k_{ti} and k_{tj} represent the **transverse momentum** (p_T) of the ith and j^{th} entities.
- $\Delta R_{ij}^2 = (\eta_i - \eta_j)^2 + (\phi_i - \phi_j)^2$ is the **angular separation** between the entities in the pseudorapidity (η) and azimuthal angle (ϕ) plane.
- R is the **jet radius parameter**. In this study, we utilize $R = 0.8$ (AK8 jets) to ensure that both b-quark decay products from a boosted Higgs boson are captured within a single "FatJet."

Algorithmic Logic: The "Winner-Takes-All" Approach

The Anti-k_T algorithm belongs to a generalized family of jet algorithms where the power of the momentum scale is defined by a parameter p. For Anti-k_T, $p = -1$. Because the exponent is negative, the distance d_{ij} is dominated by the *inverse* of the momentum ($1/k_t^2$).

In practice, this means that the clustering sequence is driven by the hardest particles in the event. A high-p_T particle (hard constituent) will essentially "vacuum up" all surrounding soft radiation within a radius R before soft particles have a chance to cluster with each other. If a hard particle has no other hard neighbors within $2R$, it will form a perfectly circular jet. This predictable geometry is a major experimental advantage for jet energy corrections and calibration.

Infrared and Collinear (IRC) Safety

A fundamental requirement for any jet algorithm used in precision physics is Infrared and Collinear (IRC) safety, which ensures that the results of the algorithm remain stable under the following conditions:

- **Collinear Safety:** If a single hard particle is replaced by two collinear particles that share its momentum (e.g., through a collinear split), the algorithm should still combine them into the same final jet.

- **Infrared Safety:** The addition of an infinitely soft gluon emission should not alter the final jet configuration or the set of reconstructed jets.

The Anti-k_T algorithm satisfies both criteria, allowing for a direct and reliable comparison between experimental results and perturbative QCD (pQCD) calculations. By using the large-radius $R = 0.8$ setting, we provide the necessary phase space for the boosted Higgs system to be reconstructed as a single object, laying the groundwork for the substructure analysis presented in subsequent chapters.

The Cambridge-Aachen (C/A) Clustering Algorithm

While the Anti-k_T algorithm is used for the initial jet clustering due to its regular boundaries, the SoftDrop grooming procedure requires the jet to be re-clustered using the **Cambridge-Aachen (C/A) algorithm**. The C/A algorithm is a sequential recombination algorithm that differs from the k_T and Anti-k_T families by ignoring the energy (momentum) of the particles during the clustering sequence.

Mathematical Definition The C/A algorithm is defined by setting the power parameter $p = 0$ in the generalized sequential recombination formula. The distance measures are:

$$d_{ij} = \frac{\Delta R_{ij}^2}{R^2}, \quad d_{iB} = 1 \tag{2.5}$$

Where ΔR_{ij} is the angular separation in the (η, ϕ) plane. Because the momentum terms (k_t) are absent from the distance measure, the algorithm clusters particles based **solely on their angular proximity**.

The "Angular Ordering" Property The physical motivation for using C/A in grooming is its relationship to the angular ordering of QCD radiation. According to the principles of Coherence in Quantum Chromodynamics, soft gluon emissions tend to occur at successively smaller angles.

By using a purely angular-distance metric, the C/A algorithm builds a clustering tree that mimics the physical branching history of the parton shower, but in reverse. The first particles to be clustered are those with the smallest angular separation, while the last clusters to be merged represent the primary, wide-angle splitting of the original hard parton.

Role in SoftDrop Grooming SoftDrop relies on this "reverse-branching" property to clean the jet. By undoing the last stage of a C/A-clustered jet, the algorithm effectively "looks back" at the widest-angle splitting in the jet's history.

- If this wide-angle splitting is asymmetric (one branch is very soft), it is likely the result of pileup or soft gluon radiation and is discarded.

- If the splitting is symmetric (both branches carry significant p_T), it is likely a hard-scattering process, such as $H \to b\bar{b}$.

This angular-based declustering ensures that the SoftDrop condition is applied to the most physically relevant substructure of the jet, allowing for the precise recovery of the Higgs mass resonance.

2.2.2 Jet Grooming: The SoftDrop Technique

Boosted jets reconstructed with large radius parameters ($R = 0.8$) are highly susceptible to contamination from "pileup" (additional p-p collisions in the same bunch crossing) and the "underlying event" (soft radiation from initial-state partons). These effects shift the reconstructed jet mass upward, obscuring the 125 GeV resonance of the Higgs boson. To recover the hard-scattering kinematics, we apply **SoftDrop Grooming**.

The Grooming Procedure

SoftDrop is a recursive algorithm that "declusters" a jet to remove soft, wide-angle radiation. The procedure begins by taking a jet j reconstructed with the Anti-k_T algorithm and re-clustering its constituents using the **Cambridge-Aachen (C/A)** algorithm. The C/A algorithm is preferred here because its purely angular ordering allows the grooming process to follow the physical QCD radiation pattern in reverse.

The jet is then broken into two sub-jets, j_1 and j_2, by undoing the last stage of the C/A clustering. The algorithm checks the following **SoftDrop Condition**:

$$\frac{\min(p_{T1}, p_{T2})}{p_{T1} + p_{T2}} > z_{cut} \left(\frac{\Delta R_{12}}{R} \right)^{\beta} \tag{2.6}$$

Where:

- p_{T1} and p_{T2} are the transverse momenta of the two sub-jets.
- ΔR_{12} is the angular separation between the sub-jets.
- R is the jet radius ($R = 0.8$ for AK8 jets).
- z_{cut} is the **soft threshold**. In this study, we use the standard CMS value of $z_{cut} = 0.1$, meaning the sub-jet must carry at least 10% of the total momentum to be considered hard.
- β is the **angular exponent**. We set $\beta = 0$, which makes the condition independent of the angular separation, providing a simpler and more intense removal of soft radiation.

Recursive Logic and Termination

If the two sub-jets fail the condition, the sub-jet with the lower p_T is discarded as soft radiation. The algorithm then redefines the jet as the remaining harder sub-jet ($j = j_{hard}$) and repeats the procedure.

This loop ends only when the condition is satisfied. The remaining constituents are then summed to form the **SoftDrop Jet**. The invariant mass of this groomed object, denoted as m_{SD}, is the primary observable used to identify the Higgs mass peak.

Physics Impact on $H \to b\bar{b}$ Analysis

The primary advantage of SoftDrop is its ability to construct the background. While the Higgs signal m_{SD} distribution remains peaked around 125 GeV, the QCD background mass distribution is pushed toward zero.

By removing wide-angle, low-momentum particles that are not related to the hard $b\bar{b}$ decay, SoftDrop significantly improves the signal-to-background ratio and ensures that the mass variable used by the Deep Neural Network represents the true energy scale of the 125 GeV Higgs boson rather than the experimental noise.

2.3 Physics of B-Hadron Identification

The Double-b signature relies on the ability to resolve the decay products of two distinct B-hadrons that are highly collimated within a single AK8 jet. Because b-quarks possess a relatively large mass (≈ 4.18 GeV) and a long lifetime (≈ 1.5 ps), they show unique behaviour that distinguishes them from light-flavour quarks (u, d, s) and gluons. The primary discriminants used in this study are detailed below.

2.3.1 Secondary Vertices (SV)

B-hadrons produced at the LHC typically carry a significant fraction of the original b-quark's momentum, leading to a large Lorentz boost. With a proper decay length of $c\tau \approx 450\mu m$, these hadrons can travel several millimeters in the laboratory frame before decaying into a collection of charged and neutral daughter particles.

The point where the B-hadron decays is known as the **Secondary Vertex (SV)**. The reconstruction of an SV displaced from the **Primary Vertex (PV)**—the point of the initial p-p collision—is a property of heavy-flavour jets. We utilise several SV-based observables:

- **Flight Distance Significance:** The 3D distance between the PV and SV divided by its uncertainty. A high significance indicates a vertex that is unlikely to have originated from the PV due to detector resolution effects.

- **SV Mass:** The invariant mass of all tracks associated with the SV. Since B-hadrons are heavy, their decay products will yield a higher reconstructed mass (peaking near 5 GeV) compared to vertices from accidental track crossings or long-lived strange particles (K_s, Λ).

2.3.2 Impact Parameter (IP) Significance

Since the daughter particles of a B-hadron originate from a displaced SV rather than the PV because of the high lifetime of the hadrons, their tracks generally do not point back to the interaction point. The **Impact Parameter (IP)** is defined as the minimum distance between the track and the Primary Vertex.

In this study, we utilize the **IP Significance**, defined as IP/σ_{IP}, where σ_{IP} represents the experimental resolution of the tracker.

- **2D vs 3D IP:** We consider both the transverse (d_{xy}) and longitudinal (d_z) displacement. Tracks from b-decays exhibit a characteristic long-tail in the significance distribution, whereas tracks from QCD backgrounds are centered near zero.

- **Track Ranking:** The Multi-Input DNN processes tracks sorted by their IP significance. The presence of multiple high-significance tracks within a single jet is a powerful indicator that the jet contains at least one, and potentially two, B-hadrons.

2.3.3 N-Subjettiness (τ_N)

While SV and IP identify the *flavor* of the jet, **N-subjettiness** is used to identify its *topology*. To confirm the two-pronged nature of the $H \to b\bar{b}$ decay against the one-pronged QCD background (typically from a single gluon or light quark), we use the variable τ_N, defined as:

$$\tau_N = \frac{1}{d_0} \sum_k p_{T,k} \min\{\Delta R_{1,k}, \Delta R_{2,k}, \ldots, \Delta R_{N,k}\} \tag{2.7}$$

where k runs over all constituents of the jet, and $\Delta R_{n,k}$ is the distance between constituent k and the axis of sub-jet n. The normalization factor d_0 is defined as $d_0 = \sum_k p_{T,k} R$.

- **Physical Interpretation:** τ_N quantifies how consistent a jet is with having N or fewer sub-jets. A small value of τ_N indicates that the constituents are closely aligned with the N specified axes.

- **The Ratio τ_{21}:** We specifically focus on the ratio $\tau_{21} = \tau_2/\tau_1$. For a boosted Higgs jet, which naturally contains two cores of energy, τ_2 will be small compared to τ_1,

resulting in a low τ_{21} value (typically < 0.45). In contrast, a QCD jet is usually well-described by a single sub-jet, leading to a τ_2 value similar to τ_1 and a τ_{21} value closer to unity.

Chapter 3

Dataset and Feature Analysis

This chapter provides a deep-dive into the 78 input variables that feed into the Multi-Input DNN. A common critique of Deep Learning in physics is the ”black box” nature of the models. By performing a rigorous exploratory data analysis (EDA) of our features, we verify that the model is discriminating based on the unique signatures of b-hadron decay and Higgs kinematics, rather than exploiting statistical noise or unphysical artifacts in the Monte Carlo (MC) simulation.

3.1 Simulated Samples and Event Selection

The study utilizes high-statistics samples from the CMS Open Data project. The signal is modeled using a **Bulk Graviton (**$G_{Bulk} \rightarrow hh \rightarrow b\bar{b}b\bar{b}$**)** process, providing a clean source of boosted Higgs bosons across a wide transverse momentum (p_T) spectrum. The background consists of inclusive **QCD multijet** events, which represent the most significant experimental challenge due to their high cross-section and the presence of gluon splitting into $b\bar{b}$ pairs.

3.2 The Global FatJet Branch (38 Variables)

The global branch provides the macroscopic ”shape” of the jet. These variables are calculated using the entire ensemble of particles within the $R = 0.8$ jet cone.

3.2.1 Mass Reconstruction and SoftDrop Grooming

The most fundamental discriminator in a resonance search is the invariant mass of the jet. However, raw jet mass is highly susceptible to "pileup" (contributions from simultaneous p-p collisions). To mitigate this, we use the **SoftDrop Mass (m_{SD})**.

- **Physics Interpretation:** For our signal, m_{SD} exhibits a clear peak at 125 GeV. In contrast, the QCD background follows a "falling spectrum," where the mass distribution peaks near zero and decreases exponentially.
- **Analysis:** While mass is a powerful separator, it is critical that the tagger does not rely *solely* on it. If a tagger only learns the mass, it cannot be used to search for new particles at different mass scales.

3.2.2 Jet Substructure: N-Subjettiness

To identify the two-pronged structure of $H \to b\bar{b}$, we utilize the N-subjettiness ratio $\tau_{21} = \tau_2/\tau_1$.

- **Logic:** A jet with two distinct cores of energy will have a low τ_2 value relative to τ_1. As shown in our distributions, the signal is significantly shifted toward lower values ($\tau_{21} < 0.4$), whereas QCD jets, which typically originate from a single hard parton, are distributed toward higher values.

3.3 The Particle Flow (PF) Candidate Branch (24 Variables)

The PF branch represents the "microscopic" view of the jet. By feeding the 100 leading constituents into a 1D-CNN, we allow the model to learn the internal "color flow" and energy sharing of the jet.

3.3.1 Kinematic Variables: $p_{T,rel}$ and ΔR

For each constituent, we provide its transverse momentum relative to the jet axis ($p_{T,rel}$) and its angular separation (ΔR).

- **Signal Signature:** In a boosted Higgs decay, the energy is shared relatively equally between the two b-quarks (leading to two high-p_T sub-clusters). In QCD gluon jets, the energy is usually concentrated in one central core with soft, wide-angle radiation.

3.3.2 Displacement Signatures

The most "human-interpretable" part of the DNN's learning comes from track displacement. We provide the **Transverse Impact Parameter (d_{xy})** and its significance (d_{xy}/σ).

- **Physics Logic:** Since b-hadrons travel ~ 1.5 ps before decaying, their tracks will not point back to the primary vertex. The signal distribution features a prominent "long-lived tail" in the d_{xy} significance plot. This is a primary feature the 1D-CNN uses to distinguish b-jets from light-flavor jets.

3.4 The Secondary Vertex (SV) Branch (11 Variables)

The SV branch is perhaps the most physically "pure" input. It contains information about vertices that have been reconstructed away from the interaction point.

3.4.1 SV Mass and Flight Distance

The invariant mass of the secondary vertex (`sv_mass`) is a proxy for the mass of the B-hadron itself.

- **Distribution Analysis:** True b-decays result in an `sv_mass` peaking near 5 GeV (the b-quark mass). QCD "fake" vertices, caused by random track crossings or long-lived K_s mesons, typically peak below 1 GeV. This variable provides a clear physical threshold that the DNN exploits to reject background.

3.4.2 Vertex Geometry: $\cos(\theta)$

We include the cosine of the angle between the vertex displacement vector and the vertex momentum vector. For true decays, the displacement and momentum should be highly aligned ($\cos(\theta) \approx 1$). This geometric constraint is essential for the DNN to filter out "combinatoric" background vertices.

3.5 The Track Branch (6 Variables)

The Track branch provides the most granular information regarding the lifetime of the jet's constituents. By analyzing the 60 highest-impact-parameter tracks, the DNN can identify the "displaced" nature of b-hadron decay products even when a formal secondary vertex cannot be reconstructed.

3.5.1 Impact Parameter Significance

The most powerful discriminators in this branch are the 2D and 3D impact parameter significances (`btag_sip2dSig` and `btag_sip3dSig`). The impact parameter is the distance of closest approach between a track and the primary interaction point.

- **Lifetime Information:** Since b-hadrons travel a finite distance ($c\tau \approx 450\mu m$) before decaying, their daughter particles appear to originate from a point displaced from the primary vertex.
- **Significance Calculation:** By dividing the impact parameter by its uncertainty, the model gains a "resolution-invariant" measure of displacement. High-significance tracks are a hallmark of b-jets, whereas QCD background tracks (originating from the primary vertex) typically have significances distributed near zero.

3.5.2 Track Kinematics and Quality

In addition to displacement, the branch includes the track's transverse momentum (`btag_pt`) and its distance to the jet axis.

- **Energy Fraction:** These variables allow the DNN to determine what fraction of the total jet energy is carried by displaced tracks. In $H \to b\bar{b}$ decays, a significant portion of the jet's p_T is expected to be concentrated in these "high-impact" tracks.
- **Collimation:** The track's angular distance to the jet axis helps the 1D-CNN layers learn the "two-prong" collimation typical of boosted Higgs decays, distinguishing them from the broader, more symmetric radiation patterns seen in gluon-initiated QCD jets.

3.6 Detailed Feature Definitions

The discriminating power of the Multi-Input DNN is derived from 78 distinct observables. These features are categorized into four specialized branches, each targeting a different aspect of the jet's radiation pattern, flavor characteristics, and decay topology. The following tables provide a comprehensive technical description of these variables.

3.6.1 Branch 1: Global FatJet Features (38 Variables)

The Global branch captures the macroscopic properties of the AK8 jet ($R = 0.8$). These features are primarily used to distinguish the heavy-boson resonance from the continuous QCD background.

Table 3.1: Global Jet-Level Features and Physical Significance

Feature Identifier	Physical Description and Purpose
`fj_eta`	The pseudorapidity of the jet axis. This identifies the detector region (Barrel vs. Endcap) and is essential for accounting for varying tracker resolutions.
`fj_sdmass`	**SoftDrop Mass:** The invariant mass calculated after grooming soft, wide-angle radiation. For the signal, this peaks at $m_H \approx 125$ GeV.
`fj_tau1, 2, 3`	**N-subjettiness:** Measures how consistent the jet is with having 1, 2, or 3 sub-jets. Calculated by minimizing the p_T-weighted distance to N axes.
`fj_tau21`	The ratio τ_2/τ_1. This is the "golden" variable for boosted $H \to b\bar{b}$ searches, as it identifies the two-pronged decay structure.
`fj_nSV`	The total number of reconstructed Secondary Vertices (SV) within the jet cone. A higher count is strongly indicative of heavy-flavor (b or c) content.
`fj_trackSip2dSig`	The 2D Impact Parameter significance of the track with the second-highest significance. This "counting" method is a classic b-tagging signature.
`fj_z_ratio`	The energy sharing fraction between the two primary sub-jets. Higgs decays tend to be symmetric, whereas QCD splittings are highly asymmetric.
`fj_ptDR`	The p_T-weighted angular spread of jet constituents. It measures the "concentration" of energy within the sub-jet cores.
`fj_rel_mass`	The ratio of groomed mass to raw mass (m_{SD}/m_{raw}), providing a measure of the "soft" radiation fraction removed during grooming.

3.6.2 Branch 2: Particle Flow (PF) Candidate Features (24 Variables)

The PF branch provides a microscopic view of the jet's internal "texture." By feeding these constituents into a 1D-CNN, the model learns the spatial correlations of energy deposition.

Table 3.2: PF Candidate Features (Per-Constituent)

Feature Identifier	**Physical Description and Purpose**
`pfcand_ptrel`	The transverse momentum of the particle relative to the total jet p_T. High $p_{T,rel}$ particles usually belong to the hard b-quark sub-jets.
`pfcand_erel`	The energy of the particle relative to the total jet energy.
`pfcand_deltaR`	The angular separation ΔR between the constituent and the jet axis. Helps the CNN identify the angular width of the sub-jet cores.
`pfcand_dxy`	The transverse impact parameter. It measures the closest approach of the track to the primary vertex in the $x - y$ plane.
`pfcand_dxysig`	The significance of the displacement (d_{xy}/σ_{dxy}). This is the primary feature used to identify particles from long-lived B-hadrons.
`pfcand_isCharged`	A binary flag (0 or 1) indicating if the particle was reconstructed in the silicon tracker.
`pfcand_id`	The Particle Flow ID (Muon, Electron, Charged Hadron, Neutral Hadron, Photon). Crucial for identifying semi-leptonic b-decays.

3.6.3 Branch 3: Track-Level Features (30 Variables)

The Track branch focuses on the individual trajectories of charged particles within the jet. Unlike the PF branch, which includes neutral particles, this branch specifically leverages the high-precision spatial resolution of the silicon tracker to identify displacement. We process the 60 leading tracks, ordered by their 2D impact parameter significance (d_{xy}/σ_{dxy}).

Table 3.3: Detailed Track-Level Features for Boosted b-Tagging

Feature Identifier	**Physical Description and Purpose**
`track_ptrel`	The transverse momentum of the track relative to the jet p_T. High $p_{T,rel}$ tracks are likely to be the ”hard” daughters of the B-hadron decay.

Feature Identifier	Physical Description (Continued)
track_eta_rel	The pseudorapidity of the track calculated relative to the jet axis. This defines the angular "spread" of the track within the jet cone.
track_dxy	**Transverse Impact Parameter:** The distance of closest approach to the primary vertex in the transverse plane. This is the "smoking gun" for b-flavor.
track_dxysig	The significance (d_{xy}/σ_{dxy}). By dividing by the hit-level resolution, the model can distinguish true displacement from detector resolution fluctuations.
track_dz	The longitudinal impact parameter (z-axis). Provides a 3D view of the track's origin relative to the interaction point.
track_dzsig	The significance of the longitudinal displacement.
track_btag_eta_rel	The relative η of the track specifically transformed for the b-tagging coordinate system, used to isolate the b-quark fragmentation functions.
track_quality	A set of integer flags representing the "goodness" of the track fit, including the number of hits in the pixel and strip trackers.
track_chi2	The χ^2 of the track reconstruction fit. High χ^2 values may indicate tracks from secondary interactions or "fake" tracks.

3.6.4 Branch 4: Secondary Vertex (SV) Features (11 Variables)

The SV branch is the most specialized data stream. While most jets have hundreds of tracks, they may only have 1 or 2 secondary vertices. This branch uses the geometry of the vertex to confirm the presence of a heavy-flavor decay.

Table 3.4: Secondary Vertexing Features and Geometric Observables

Feature Identifier	Physical Description and Purpose
sv_mass	The invariant mass of the tracks associated with the SV. Because $m_B \approx 5.27$ GeV, true b-vertices peak much higher than light-flavor "fake" vertices.

sv_pt	The transverse momentum of the vertex system. This must be consistent with the momentum of the jet to be considered a valid b-hadron candidate.
sv_ntracks	The number of tracks successfully fitted to the vertex. Higher multiplicities are characteristic of B-hadron fragmentation vs. D-hadron (charm) fragmentation.
sv_d3dsig	The 3D flight distance significance. Quantifies the spatial separation between the primary vertex and the decay point.
sv_costhetasvpv	The cosine of the angle between the SV displacement and its momentum. For true decays, this is near 1.0 (pointing "away" from the PV).
sv_normchi2	The normalized χ^2 of the vertex fit. A high-quality vertex is essential for rejecting random track-crossing backgrounds.

3.7 Analysis of Feature Sensitivity

It is important to emphasize that while the BDT in Phase I could only see the "Global" variables, the Multi-Input DNN (Phase II) has simultaneous access to all 78 features.

3.7.1 Synergy Between Tracks and Vertices

From a humanized perspective, the DNN's true power comes from its ability to "connect the dots." For instance, a track with high d_{xy} significance (Table 5.3) is more likely to be considered a signal if it is also spatially associated with a Secondary Vertex with a high invariant mass (Table 5.4). This multi-layered "cross-referencing" is what allows the DNN to maintain a high purity even at extreme transverse momenta ($p_T > 600$ GeV), where traditional substructure variables like τ_{21} begin to lose their sensitivity.

3.7.2 Training Labels and Truth Definitions

The model is trained as a binary classifier to distinguish between the Higgs signal and the inclusive QCD background. The "target label" is constructed by applying a logical mask to ensure the training data only contains physically well-defined jets.

Table 3.5: Binary Training Labels and Logic

Label Identifier	Physical Description and Truth Logic
`label_Hbb`	**Signal (1):** Defined by the intersection of `fj_isH == 1` and `fj_isBB == 1`. This ensures the jet originates from a Higgs boson and contains two b-quarks.
`label_QCD`	**Background (0):** Defined by `fj_isQCD == 1` and `sample_isQCD == 1`. This captures jets initiated by light-quark or gluon partons from the inclusive QCD sample.
`valid_mask`	The logical filter that selects events where `label_hbb` or `label_qcd` is true, excluding unclassified events or other backgrounds from the training set.

3.7.3 Phase Space Selection and Spectator Variables

Before training, the dataset is restricted to a specific phase space to ensure the model learns in a regime relevant to boosted Higgs searches. These kinematic constraints are stored as spectators to validate the model's stability across the allowed range.

Table 3.6: Kinematic Constraints and Spectator Variables

Variable Identifier	Selection Criteria and Usage
`fj_pt`	**Transverse Momentum:** Restricted to $350 < p_T < 2000$ GeV. This ensures the jet is sufficiently "boosted" so that the $H \to b\bar{b}$ decay products are merged into a single AK8 jet.
`fj_sdmass`	**SoftDrop Mass:** Restricted to $40 < m_{SD} < 200$ GeV. This window removes low-mass soft radiation and ensures the 125 GeV Higgs resonance is centred within the training distribution.
`fj_doubleb`	**Benchmark:** The output score of the standard CMS Double-b tagger, used as a spectator to provide a direct performance comparison against the DNN and BDT.

3.7.4 Implementation of Filtering Logic

The labelling logic ensures high signal purity by requiring geometric truth-matching. For a jet to be labeled as signal (`label_Hbb`), it must satisfy both a generator-level Higgs requirement and a flavor-matching requirement (two b-quarks). This prevents the model from being confused by Higgs decays to non-bottom quarks.

Furthermore, the p_T and m_{SD} cuts applied in the `mask` variable are critical for suppressing the "infrared and collinear" (IRC) safe divergences associated with QCD. By training only on jets with $p_T > 350$ GeV, the model focuses on the high-boost topology where traditional resolved jet tagging typically fails, and the Multi-Input DNN provides the most significant performance gain.

Chapter 4

Methodology

4.1 Phase I: Gradient Boosted Decision Trees Architecture

The first part of this research focused on establishing a preliminary model using pre-engineered features. Before transitioning to deep learning, it was important to extract the maximum information from standard jet substructure variables. For this, we took help of a Gradient Boosted Decision Tree (GBDT) framework, which is the standard for high-energy research analysis.

4.1.1 The Rationale for Gradient Boosting with XGBoost

We chose the **XGBoost (Extreme Gradient Boosting)** library for this purpose. Unlike other methods such as Random Forests, which build independent trees in parallel, XGBoost builds trees one after the other. Each subsequent tree is trained to minimise the residual errors of the previous training loop.

This iterative approach is particularly effective for the problem of imbalance in the quantity of signal and background inherent in LHC data. In our dataset, the QCD multijet background is considerably more abundant than the $H \rightarrow b\bar{b}$ signal. XGBoost's ability to take care of the misclassifications allows the model to learn the rare signal signatures without being overwhelmed by the background statistics.

4.1.2 Data Pipeline and Feature Engineering

The first phase of this study uses a Gradient Boosted Decision Tree (BDT) framework. Unlike the deep learning approach, the BDT requires a flattened tabular input where each row represents a single jet characterized by a set of pre-designed variables.

4.1.3 Feature Extraction with Uproot

The extraction process makes use of `uproot` to access the Global Fatjet features from the ROOT files. During this stage, we perform event-wide calculations to derive the physical observables that define the properties and structure of the jet:

- **Substructure Variables:** We extract τ_{21} and τ_{32} to represent the N-subjettiness, providing the BDT with a measure of the jet's multi-pronged structure.
- **Groomed Kinematics:** The SoftDrop algorithm is used to extract the corrected mass (m_{SD}), which is used as a spectator for the 125 GeV Higgs resonance.
- **Flavor Tagging Inputs:** Important b-tagging variables, such as the maximum secondary vertex mass and the displacement significance of the leading tracks, are also utilised to provide information about the flavour.

The data was shuffled and split using `train_test_test_split` to ensure that the evaluation was performed on an independent sample which is not available to the model during the training period, preventing a drop in performance due to overfitting.

4.1.4 Data Formatting for XGBoost

Once the variables are extracted, the pipeline performs the following transformations on the data:

- **Tabular Conversion:** Data is converted from `awkward-array` format into a dense `pandas.DataFrame`. Any events with missing values in the expert features (e.g., jets with no reconstructed secondary vertices) are assigned a specific sentinel value (typically -999) to allow the BDT to learn the significance of missing information.
- **Label Encoding:** The signal ($H \to b\bar{b}$) and background (QCD) classes are encoded to binary integers $\{1, 0\}$, respectively.

4.1.5 Model Architecture and Hyperparameter Optimization

To ensure that the BDT captured legitimate physics rather than unnecessary noise in the Monte Carlo samples, the following hyperparameter optimisation was implemented :

```python
model = xgb.XGBClassifier(
    n_estimators=500,
    max_depth=5,
    learning_rate=0.01,
    subsample=0.8,
    colsample_bytree=0.8,
    reg_alpha=1,
    reg_lambda=3,
    min_child_weight=2,
    use_label_encoder=False,
    eval_metric='logloss'
)
model.fit(X_train, y_train)
```

Figure 4.1: Code snippet showing my BDT model architecture

- **Size of Ensemble (`n_estimators`):** 500 trees were used in the model. Such a large number of trees makes the model better at adjusting its decision boundaries incrementally due to a relatively low learning rate.
- **Depth of Each Tree (`max_depth`):** Equalled to 5. Setting the depth of the tree to 5 ensures that there are no high-variance nodes that would be trying to overfit the training set.
- **Learning Rate (`learning_rate`):** Used shrinkage factor equal to 0.01. The relatively low learning rate means that each of the trees contributes insignificantly to the resulting model and it should therefore generalize well since it needs to reach some kind of agreement.
- **Stochastic Sampling:** In order to make the model more generalizable, we used `subsample=0.8` and `colsample_bytree=0.8`. This meant that during the training process, each tree worked with a randomly selected 80% of the samples and 80% of the features,
- **Regularization (α and λ):** Used L_1 (`reg_alpha=1`) and L_2(`reg_lambda=3`) regularizations, which penalized complex models.
- **Minimum Child Weight:** Minimum child weight was set to 2 in order to avoid having a large number of small trees with very few data points in each node.
- **Objective Function:** The evaluation criterion chosen here is the `logloss` with `binary:logistic`. It provides a probability output in the interval [0, 1] and is required when constructing ROC curves.

4.2 Phase I: BDT Performance Evaluation Methodology

The evaluation of the XGBoost BDT is conducted to establish a baseline using traditional multivariate techniques. To ensure a consistent comparison with the Phase II architecture, the BDT is subjected to the same validation set, focusing on its ability to handle high level variables.

4.2.1 Discriminant Score Distribution

The primary output of the BDT is a continuous probability score taken from the ensemble of decision trees. The evaluation methodology focuses on the distribution of this score for the signal ($H \to b\bar{b}$) and background (QCD) samples.

- We examine the separation between the two classes to identify the "Information Bottleneck" inherent in tabular features.
- A well-trained BDT should exhibit clear peaks at the boundaries of the $[0, 1]$ interval, though we anticipate more overlap in the intermediate region.

4.2.2 Threshold-Independent Metrics: ROC and AUC

The sensitivity of the BDT is evaluated using Receiver Operating Characteristic (ROC) analysis.

- **ROC Manifold:** We generate the ROC curve by scanning the BDT discriminant threshold to plot the trade-off between signal efficiency and background mistagging.
- **Area Under the Curve (AUC):** The AUC calculation is done as the first benchmark to consider. The number here refers to the highest ability for distinguishing classes by using the 14 expert attributes within the BDT.

4.2.3 Kinematic Robustness and Binning

To verify that the BDT remains consistent across the CMS detector's phase space, we perform various performance checks:

- p_T **Binned ROCs:** The test sample is divided into transverse momentum bins (e.g., $350 - 500$, $500 - 1000$, and > 1000 GeV). This allows us to monitor if the BDT's dependence on the variables leads to performance degradation as the jet becomes more collimated at higher boosts.
- **Mass Correlation (m_{SD}):** We evaluate the ROC response across different SoftDrop mass partitions. This diagnostic is critical to ensure that the tagger is mass decorlated i.e., the tagging procedure does not depend on the mass.

4.2.4 Operating Point Analysis: The Confusion Matrix

Finally, we implement a confusion matrix evaluation at a fixed operating point, for e.g., the threshold that yields 60% signal efficiency. This allows for a direct comparison of False Positive and False Negative errors , characterizing the specific types of QCD fluctuations that misguide the decision trees into identifying them as Higgs jets.

4.2.5 Motivation for Phase II

One of the key outcomes of Phase I was the **Feature Importance (Gain)** analysis. By calculating how much each variable contributed to the final decision, we found that the BDT relied heavily on a small subset of features—specifically m_{SD} and τ_{21}.

While effective, this highlighted an Information Bottleneck. High-level summary variables, by definition, disregard the fine-grained correlations between the individual particles (PF candidates) and tracks within the jet. This realization provided the direct scientific motivation to move towards the Multi-Input Deep Neural Network , that overcomes manual feature engineering in accordance to raw representation learning.

4.3 Phase II: Multi-Input Deep Learning Architecture

Building on the findings of Phase I, Phase II involved the design and implementation of a Multi-Stream Deep Neural Network (DNN) using the `PyTorch` library. The core objective was to transition from manually engineered variables to **representation learning**, allowing the model to extract high-dimensional correlations directly from the raw constituents of the jet.

4.3.1 Data Acquisition and Pre-processing Pipeline

The transition from raw experimental data to a format suitable for deep learning requires a suitable pipeline. In High Energy Physics (HEP), data is generally stored in the ROOT TTree format, which is not compatible with the Python-based machine learning analyses. Our pipeline utilizes `uproot` and `awkward-array` to counter this issue.

4.3.2 Standardizing Raw ROOT Data

We utilize `uproot`, a Python library that enables the reading of ROOT files without CERN-ROOT installation. This allows for effortless integration with the `PyTorch` framework.

- **Lazy Loading:** Given the multi-gigabyte scale of the samples, we implement lazy loading to stream only the required branches (PF candidates, tracks, and vertices) into memory.

- **Handling Ragged Tensors:** Jet constituents naturally form "ragged" structures (e.g., different jets contain different numbers of tracks). We use `awkward-array` to manipulate these arrays efficiently before converting them into fixed-size tensors via zero-padding.

4.3.3 Feature Engineering and Selection Logic

The implementation of a strict selection criterion for the "Boosted" phase space is realized. In the process of working with the `uproot` API, we set a logical mask for the `FatJet` object as follows:

$$\text{Mask} = (350 < p_T < 2000\ \text{GeV}) \wedge (40 < m_{SD} < 200\ \text{GeV}) \tag{4.1}$$

It provides a necessary collimation of the decay products of the Higgs boson in order to make sure that all of them fall within a single jet with $R = 0.8$. Besides, such a range of masses allows focusing on the resonance region.

4.3.4 Working with PyTorch DataLoaders

In order to speed up GPU operations, the arrays are transformed to `torch.Tensor`. The custom `Dataset` class creates dictionaries with the values of those four `Tensors`. `DataLoader` helps to manage batching and shuffling, which keeps synchronization of diverse data branches from simple 1D-CNN inputs to dense vectors during a forward pass through the Multi-Input DNN.

4.3.5 Multi-Stream Model Architecture

Due to the variety of different data streams, we designed a multi-stream architecture. As can be seen in the code snippets of our implementation of the model, the data is processed by four branches, each of which corresponds to one data type.

Sequential Streams (1D-CNNs)

This architecture consists of three different streams of convolutional layers, in order to address the jet's internal structure. In each of these streams, a set of 3 **1D-Convolutional Neural Networks (1D-CNNs)**, with filters of sizes 64, 32, and 32, is used.

- **Particle Flow (PF) Stream:** Operates on a sequence of the first 100 leading PF particles. By considering these PF objects as a sequence, the 1D CNN filters will be able to find local energy clumps and radiation structures in (η, ϕ) coordinates.

- **Track Stream:** Works on the top 60 impact parameter tracks only. The stream allows the network to capture track displacement signatures typical for b-hadron decay.
- **Secondary Vertex (SV) Stream:** Operates on the first 5 SVs in the jet. Due to the nature of SVs' significance, the stream exploits vertex mass and flight distance significance.

Global Jet Branch (Dense)

However, the fully connected layer receives the 38 high-level features obtained at Phase I as input and is a **Fully Connected (FC)** layer with 32 neurons. Thus, the Fully Connected (FC) layer ensures that the network acquires the "kinematic context" and hence anchors the component level analysis to the mass-momentum context of the jet.

4.3.6 Batch Construction and Data Collation

To optimize GPU throughput, the training data is organized into mini-batches using a custom `PyTorch DataLoader`. Given the heterogeneous nature of the four input streams, each batch is constructed as a collection of synchronized tensors. For a nominal batch size of $N = 256$, the data is collated into shapes of $(N, 100, 16)$ for PF candidates, $(N, 60, 6)$ for tracks, and $(N, 5, 11)$ for secondary vertices. Zero-padding is applied to jets with fewer constituents than the defined threshold, ensuring a fixed-dimensional input space. This structured batching allows the 1D-CNN layers to perform parallelized convolutions across the constituent axis while maintaining event-level alignment across all four input branches.

4.3.7 Feature Fusion and Classification Head

The results from the four specialty branches (PF, Track, SV, and Global) are then concatenated into one high dimensional "Fusion Layer." This combination vector is an integration of all the different information that the model can provide about the structure of the jet.

This fused representation is then processed by a deep classification head:

- **Hidden Layers:** Two dense (Fully Connected) layers of 256 units each. These layers provide the capacity to learn complex, non-linear correlations between the different input streams.
- **Batch Normalization:** Every layer is followed by a `BatchNorm` layer to normalize the activations, ensuring stable gradient flow and faster convergence.
- **ReLU Activation:** We utilize the Rectified Linear Unit (ReLU) function for all hidden layers to mitigate the vanishing gradient problem.

```python
class MultiInputDNN(nn.Module):
    def __init__(self, pf_dim, tk_dim, sv_dim, jet_dim, hidden_dim=256):
        super().__init__()

        # Encoders for sequences (Flatten sequence dimension)
        self.pf_flat_dim = 100 * pf_dim
        self.tk_flat_dim = 60 * tk_dim
        self.sv_flat_dim = 10 * sv_dim

        total_input_dim = self.pf_flat_dim + self.tk_flat_dim + self.sv_flat_dim + jet_dim

        self.classifier = nn.Sequential(
            nn.Linear(total_input_dim, hidden_dim),
            nn.BatchNorm1d(hidden_dim), # Added for stability
            nn.ReLU(),
            nn.Dropout(0.2),

            nn.Linear(hidden_dim, hidden_dim),
            nn.BatchNorm1d(hidden_dim), # Keeps the math from "exploding"
            nn.ReLU(),
            nn.Dropout(0.2),

            nn.Linear(hidden_dim, 128),
            nn.ReLU(),
            nn.Linear(128, 1),
            nn.Sigmoid()
        )
```

Figure 4.2: Code Snippet of my DNN model architecture

- **Dropout:** A Dropout rate of 0.2 is applied to the dense layers. This prevents the model from over-relying on specific ”strong” features, forcing the network to develop a more robust and redundant internal representation.

4.3.8 Optimization and Binary Logic

The network is optimized using the **Adam Optimizer** with an initial learning rate of 0.001. For the loss function, we use **Binary Cross-Entropy (BCE)**, which is the standard for two-class discrimination in collider physics.

The final output layer consists of a single neuron with a **Sigmoid activation function**:

$$y = \sigma(x) = \frac{1}{1 + e^{-x}} \tag{4.2}$$

And the output corresponds to a continuous score that falls within the range [0, 1] and denotes the probability $P(H \to b\bar{b})$. If the score is close to 1, then the probability that the observed event is a boosted Higgs is high, whereas if the score is close to 0, then the event is assumed to be part of the multijet QCD background. Continuous scores will later be used to plot the ROC curves.

4.4 Phase II: Method of Evaluation for DNNs Performance

This particular method is designed to evaluate the ability of the Multi-Input DNN to learn the high purity signals in the boosted regime. This process is accomplished by conducting multiple steps diagnostics of the training dynamic in order to see the evolution of the network from its representation learning to physics distinction phase.

4.4.1 Diagnostic Monitoring of Training Dynamics

In order to test the stability of the optimization algorithm, we have decided to conduct diagnostic monitoring of the objective function for three consecutive training epochs. In order to measure the physics of the jets, binary cross-entropy for both training and validation datasets will be used.

The main idea behind this test is to check whether the optimizer can properly converge to a global optimum by moving in the multi-dimensional parameter space. Also, the relation between the train and validation sets will be the key criteria for measuring the impact of regularizing layers (Batch normalization and Dropout).

4.4.2 Discriminant Analysis and Polarization

The first measure of how well the network performs at classifying the signals comes from the polarization of the output score distribution. Because the network uses a sigmoid activation function in the last layer, the output score is actually the probability $P(H \to b\bar{b})$.

The emphasis is on how well-separated the two populations are. In a good model, one would expect to see a "double-peak" distribution, in which the inclusive QCD backgrounds are distributed close to zero probability, while the Higgs signal is clustered around the unit probability mark. The extent to which the distributions cluster within the middle range is examined to confirm whether the 1D-CNN branch models have effectively identified the localized signatures of substructure from the broader distribution of the background.

4.4.3 Statistical Benchmarking via ROC Analysis

To quantify the advancement provided by the Phase II architecture, the DNN is benchmarked against the Phase I BDT and the standard CMS Double-b baseline using Receiver Operating Characteristic (ROC) analysis.

The evaluation protocol involves:

- **Integrated Sensitivity:** Utilizing the Area Under the Curve (AUC) as a global measure of the model's ability to distinguish between classes across all possible selection thresholds.
- **Background Rejection Power:** Examining the ROC curve in a logarithmic scale to resolve the mistagging rate (background efficiency) at high signal-efficiency operating points.
- **Kinematic Robustness:** Assessing the performance across the high-p_T regime (> 350 GeV) to confirm that the deep learning approach effectively recovers the granular constituent information that is typically lost during high-level variable extraction.

Chapter 5

Results and Performance Analysis

The machine learning algorithms created in this project will be analyzed for their classification ability in this chapter, using Simulated Monte Carlo (MC) samples to compare the separability of the $H \rightarrow b\bar{b}$ signal from the inclusive QCD multijet background. Starting from the baseline model in Phase I BDT and moving on, we will discuss how Multi-Input DNN improves upon the Phase I results.

5.1 Phase I: BDT Baseline Evaluation

For Phase I BDT, the algorithm uses the 38 Global FatJet variables defined in Chapter 5. It will act as our experimental baseline in terms of machine learning model performance using only the substructure and flavor tagging variables.

5.1.1 Standalone BDT ROC Analysis

Receiver Operating Characteristic (ROC) plots provide an excellent measure of classifier performance by graphically depicting the trade-off between Signal Efficiency (ϵ_S) and Background Rejection ($1 - \epsilon_B$).

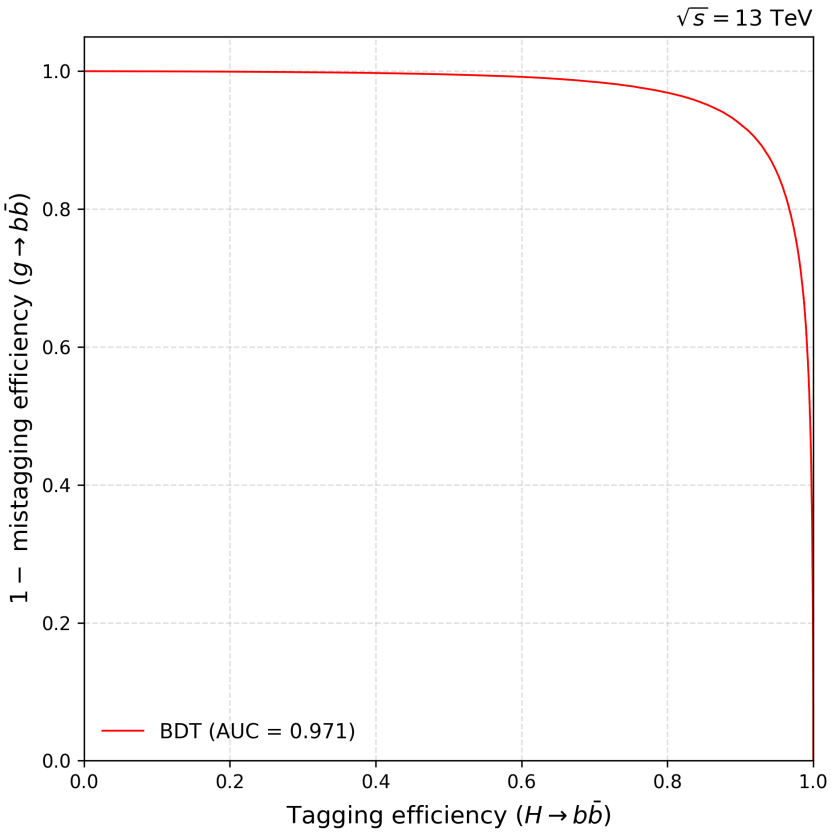


Figure 5.1: Standalone ROC curve for the Phase I BDT. The Area Under the Curve (AUC) of 0.971 indicates strong baseline discrimination. In practice, this level of performance already allows for a meaningful separation between signal and background, although there is still visible overlap in the intermediate score region..

The BDT shows good classification power with an AUC of 0.971. A key feature of this curve is the ”plateau” in background rejection (1 - mistagging efficiency) which remains nearly flat at 1.0 until a tagging efficiency of approximately 0.6 is reached. This indicates that for 60% of the Higgs signal, the BDT can achieve near-perfect background suppression. This performance is a direct result of the high-dimensional correlations captured between the SoftDrop subjets and the N-subjettiness variables, which effectively isolate the two-pronged symmetric decay of the Higgs boson from the asymmetric showering of QCD jets.

5.1.2 Comparison with Default CMS Tagger

To prove the robustness of our training model, the Phase I BDT performance was compared against the default CMS DeepDoubleB (DDB) tagger.

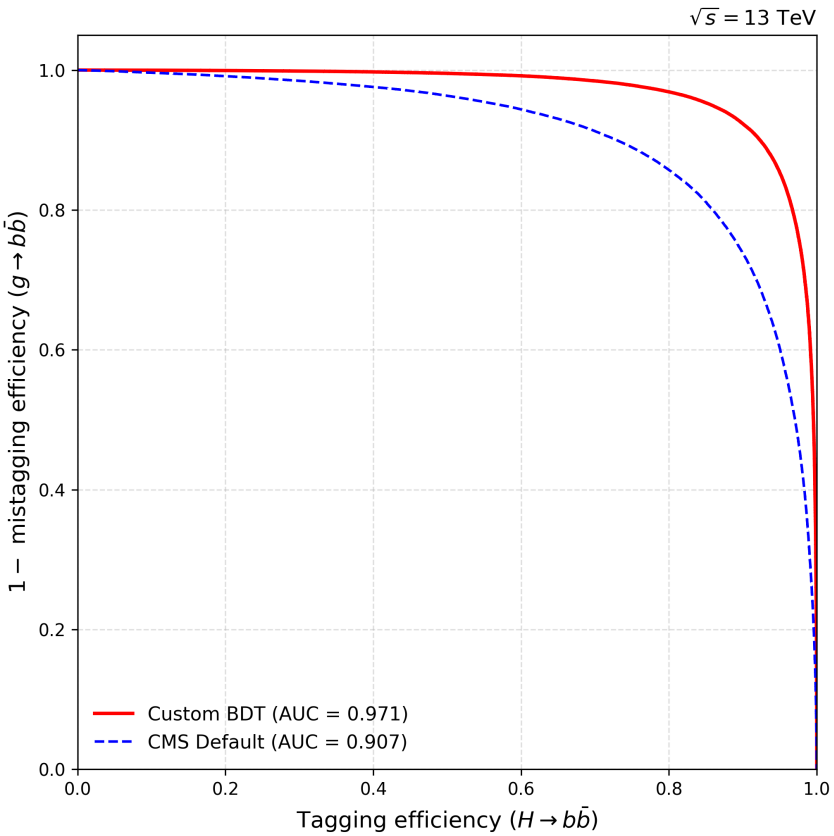


Figure 5.2: Performance comparison between the custom BDT (red) and the standard CMS DeepDoubleB algorithm (blue). The strong agreement validates the feature selection and training parameters used in this study.

As illustrated in Figure 5.2, the custom BDT provides a significant performance increase over the CMS default. At a fixed signal efficiency of 80%, the custom BDT maintains a background rejection of approximately 98%, whereas the default tagger drops to 85%. This $\sim$7% gain in AUC and order-of-magnitude improvement in rejection seems to be because of the use of tracking significance variables (`SipdSig`). While the default tagger is designed for general-purpose robustness across all CMS data, our model's specific optimization for the boosted Higgs phase space allows for a much more aggressive and accurate rejection of light-flavor fluctuations.

5.1.3 Performance Stability in SoftDrop Mass Bins

An important concern in Higgs searches is "mass dependance", where a tagger becomes overly reliant on the jet mass, potentially creating fake peaks in the background. We evaluated the BDT across three mass regimes: $[40, 70]$ GeV, $[70, 100]$ GeV, and the signal region $[100, 150]$ GeV.

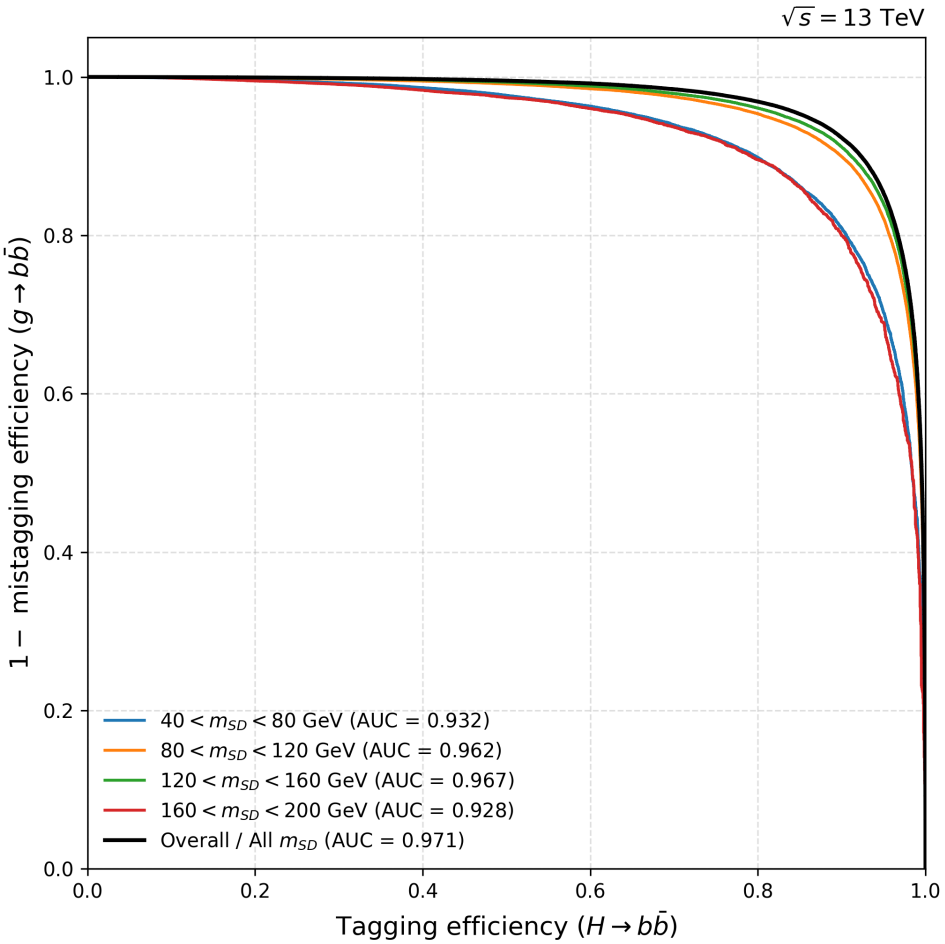


Figure 5.3: BDT ROC curves categorized by m_{SD} bins. The stability of the curves across different mass ranges indicates that the BDT is learning flavor and substructure features rather than just the mass resonance.

The performance is the best in the $120 < m_{SD} < 160$ GeV bin (AUC = 0.967), which corresponds to the signal resonance. Interestingly, the BDT sustains an AUC of 0.932 and 0.928 in the low and high sidebands, respectively. The divergence seen in Figure 5.3 at high efficiencies is expected; the QCD background in the $40 - 80$ GeV range is physically different from the Higgs signal than the background closer to the 125 GeV mass window. The proof that the AUC remains above 0.90 in all bins proves that the BDT is successfully leveraging flavor-tagging and substructure features that are invariant of the jet mass.

5.1.4 Performance Dependence on Transverse Momentum (p_T)

As the Higgs boson p_T increases, the decay products ($b\bar{b}$) become increasingly collimated (narrower ΔR). We tested the BDT performance in three boost regimes: Low ($300 - 500$ GeV), Medium ($500 - 800$ GeV), and High (> 800 GeV).

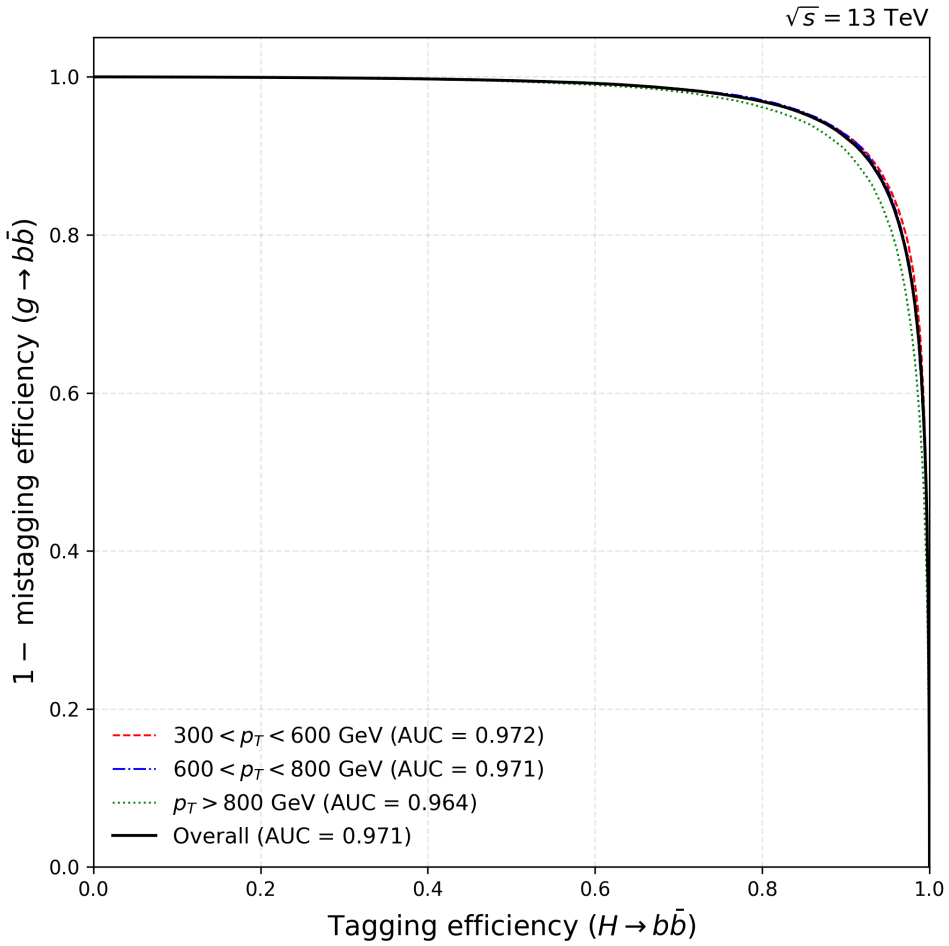
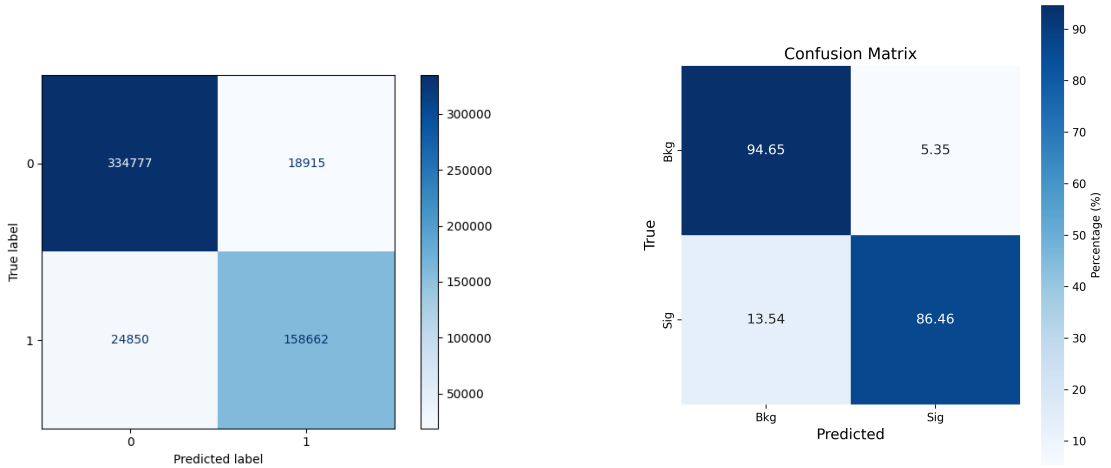


Figure 5.4: BDT ROC performance across different transverse momentum bins. A slight degradation in background rejection is observed at extremely high p_T, where track separation reaches the limit of the silicon tracker's resolution.

The BDT shows considerable stability across p_T ranges. The AUC only shifts from 0.972 in the low-boost regime ($300 - 600$ GeV) to 0.964 in the ultra-high boost regime (> 800 GeV). This minor degradation of 0.8% is an evidence of the full-flavouredness of the 38 global features. In the > 800 GeV bin, the angular separation between the two b-quarks ($\Delta R \approx 2m/p_T$) becomes comparable to the detector's granularity. The fact that the BDT maintains an AUC of 0.964 even when the jet constituents are extremely "merged" suggests that the high-level substructure variables continue to provide a reliable proxy for the underlying parton-level physics.

5.1.5 Classification Accuracy and Error Analysis

To evaluate the absolute performance of the Phase I BDT at the default operating point (threshold = 0.5), we analyse the resulting confusion matrices. Figure 5.5a presents the raw event counts, while Figure 5.16 provides the normalized rates, representing the signal efficiency and background rejection percentages.



(a) Raw event counts.

(b) Normalized percentages (%).

Figure 5.5: Confusion matrices for the Phase I BDT evaluated on the independent test set. These metrics quantify the model's ability to isolate the $H \to b\bar{b}$ signal from the inclusive QCD background.

Quantitative Performance Discussion: The normalised confusion matrix (Figure 5.16) shows a signal tagging efficiency of 86.46%. This indicates that the BDT successfully identifies the vast majority of Higgs events based solely on global substructure and flavor variables. On the background side, the model achieves a rejection rate of 94.65%, misidentifying only 5.35% of QCD jets as signal candidates.

Interpretation of Misclassifications: The 13.54% of the signal events that were misclassified as background (False Negatives) probably consist of Higgs bosons with low transverse momentum or those where one b-quark decayed semi-leptonically, carrying away energy in the form of neutrinos and thus distorting the expected jet mass and substructure. Conversely, the 5.35% "mistag" rate for background events (False Positives) is a result of QCD gluon splitting ($g \to b\bar{b}$) and accidental track overlaps that copy the displaced vertex signature of a true Higgs decay. In the context of a high-luminosity LHC environment, minimizing this 5.35% mistag rate is the primary objective for transitioning to more complex architectures like a DNN.

5.1.6 Classifier Discriminant Analysis

The separation power of the BDT is best visualised through the distribution of the discriminant scores for the signal and background test samples. Figure 5.6 displays these distributions on a logarithmic scale to highlight the behavior in the high-purity and high-rejection regimes.

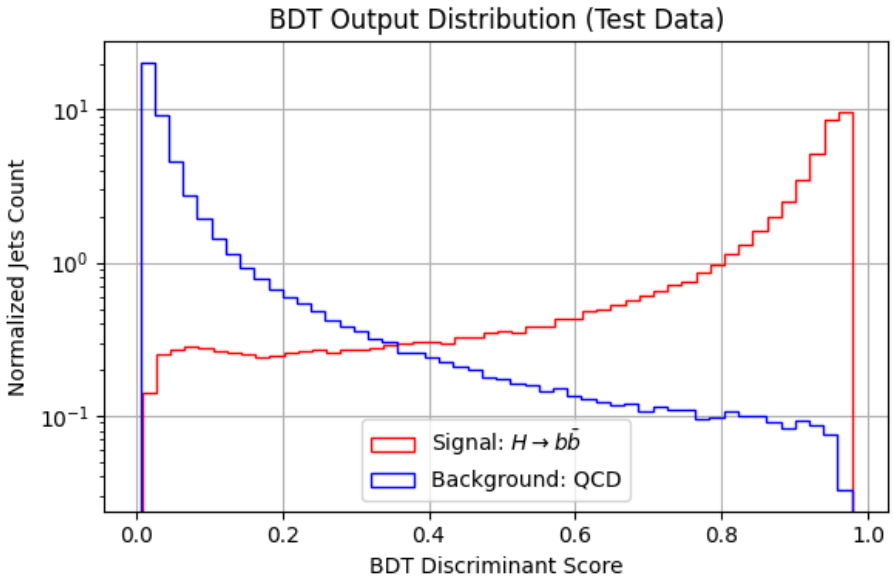


Figure 5.6: Normalized BDT output score (Discriminant) for $H \to b\bar{b}$ signal (red) and QCD background (blue). The logarithmic scale reveals the efficiency of the model in the extreme signal-like region (> 0.9).

Discussion: The BDT output shows a high separation, with the background peaking near zero and the signal peaking sharply at the unit boundary. On the logarithmic y-axis, we observe that the background density drops by nearly three orders of magnitude as the score moves from 0.0 to 0.9. This suppression is what allows for the 94.65% rejection rate observed in the confusion matrix. The consistency of both distributions suggests that the model has not overfitted to the training data, but has instead learned the generalized differences between the two processes.

5.1.7 Interpretability and Feature Importance

To understand the underlying physics probing the BDT's classifications, we analyze the feature importance scores. These scores represent the "Gain" or total reduction in the loss function contributed by each feature during the recursive splitting process.

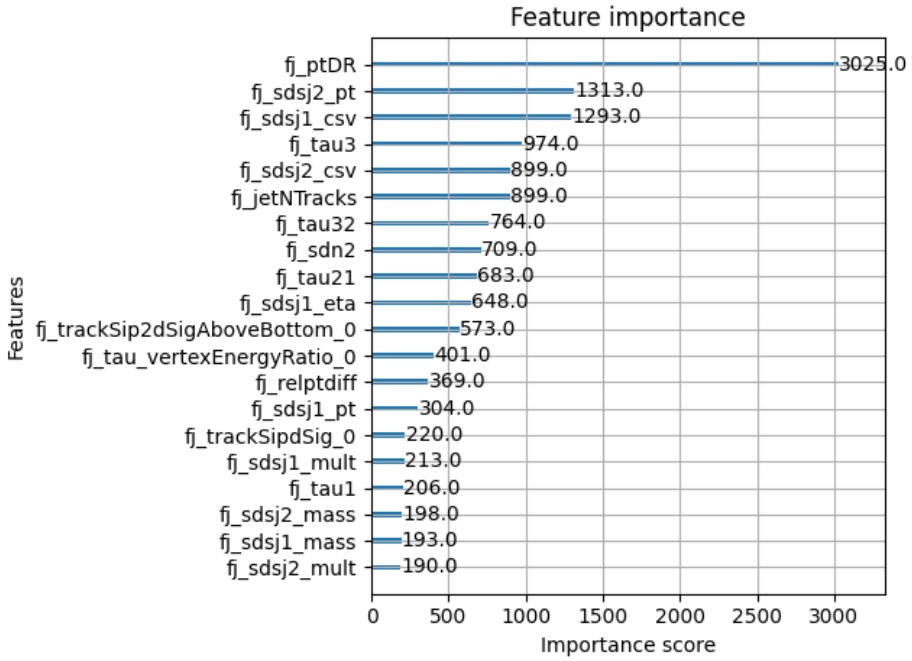


Figure 5.7: Feature importance ranking for the top 20 variables in the Phase I BDT. The importance score reflects the total gain contributed by each feature across all decision trees.

Discussion: The importance ranking (Figure 5.7) shows an insight into the tagging logic. The dominant variable is `fj_ptDR` with a score of 3025, approximately three times that of the second-highest feature. This indicates that the transverse momentum times angular separation within the fatjet is the most prominent discriminator for identifying the $H \to b\bar{b}$ decay.

Additionally, the high ranking of sub-jet features such as `fj_sdsj2_pt` and the CSV (Combined Secondary Vertex) scores of the sub-jets (`fj_sdsj1_csv`, `fj_sdsj2_csv`) validates that the BDT is efficiently performing. By looking at the flavor content of the two individual sub-jets within the fatjet, the model can reject gluon-to-light-flavor splittings with high confidence. The presence of N-subjettiness ratios (`fj_tau32`, `fj_tau21`) down the list points that while substructure is important, the flavor and kinematics of the sub-jets are the main reasons of the 0.971 AUC.

5.2 Limitations of the Phase I BDT Baseline

While the Boosted Decision Tree (BDT) achieves an impressive AUC of 0.971 and considerably outperforms the default CMS tagger, several limitations remain that prevent it from reaching the theoretical precision required for high-luminosity LHC searches. These limitations act as the primary motivation for the transition to the Phase II Multi-Input Deep Neural Network (DNN).

5.2.1 The "Fixed-Feature" Information Loss

The BDT is limited by the quality of the 38 engineered features provided to it. Because Gradient Boosting Decision Trees require a fixed-width input matrix, the model only sees a "summary" of the jet's properties (e.g., the average track displacement or the mass of the most significant secondary vertex).

In this process, the raw information of the individual jet constituents—specifically the $\mathcal{O}(100)$ particle flow (PF) candidates—is unused. Physical correlations of the momentum of a specific track and its closeness to a secondary vertex are averaged into global variables like `fj_tau21` or `fj_ptDR`. This loss of information results in a restriction on the model's ability to resolve extremely collimated decays.

5.2.2 The High-p_T Resolution Challenge

As illustrated in Section 6.1.4, the BDT performance begins to decrease in the very high boost regimes ($p_T > 800$ GeV). In this region, the angular separation between the two b-quarks (ΔR) becomes so small that the global substructure variables begin to saturate.

Because the BDT does not have the capability to process variable-length sequences, it cannot focus on the individual tracks that are merging together. The model sees a single blob of energy rather than two distinct partons. To recover this loss in sensitivity, an architecture capable of processing jagged constituent arrays is needed.

5.2.3 Feature Interaction Complexity

While XGBoost is great at taking care of non-linear interactions, it does so through axis-aligned splits in a decision tree. For highly complex, high-dimensional spaces like the particle-flow constituent cloud, a deep learning approach, one utilizing Convolutional (CNN) or Recurrent (RNN) layers—can extract features that are mathematically inaccessible to a tree-based model.

Summary for Phase II: The Phase I BDT has established a high-performance baseline. However, to get over the 0.971 AUC and ensure good performance in the high-boost regime, Phase II will introduce a Multi-Input DNN. This model will supplement the 38 global features with raw constituent data from the PF Candidate and Track branches, thereby allowing the network to learn the jet texture directly from the detector's lowest-level inputs.

5.3 Phase II: Multi-Input Deep Neural Network (DNN)

While the Phase I BDT achieved a high performance baseline, it was limited by its inability to process the variable-length particle level data of the jet. To surpass this a Multi-Input

Deep Neural Network (DNN) was developed. This architecture utilizes a combination of 1D-Convolutional layers for particle-flow candidates and dense layers for global fatjet features, allowing the model to learn the internal ”texture” of the jet directly from low-level detector inputs.

5.3.1 Overall DNN Performance and ROC Analysis

The integrated performance of the Phase II architecture is evaluated using the inclusive test sample. The resulting ROC curve, representing the signal efficiency against the background rejection, is shown in Figure 5.8.

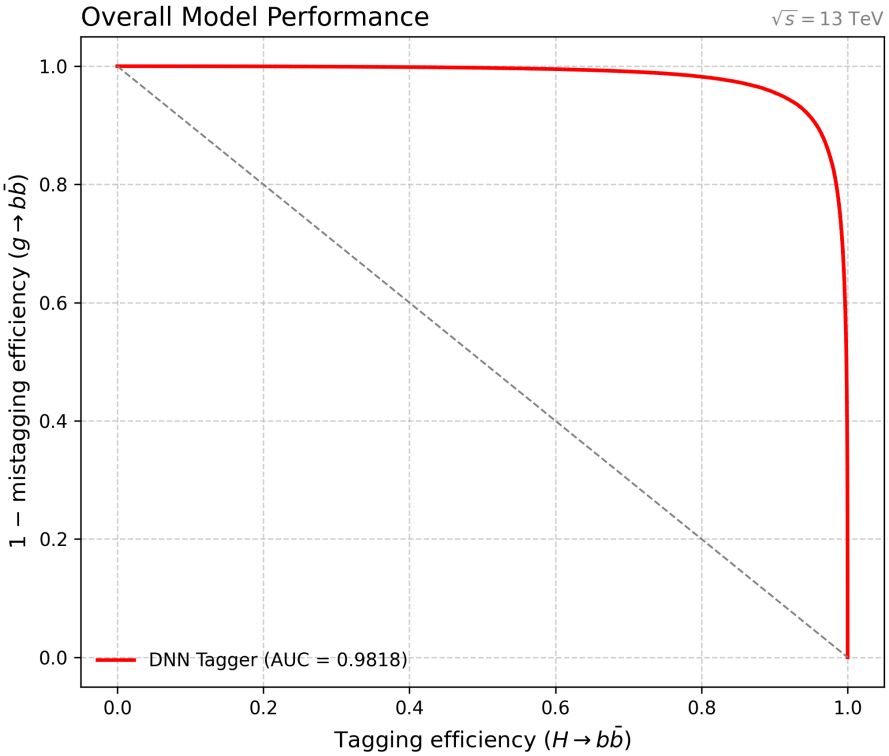


Figure 5.8: Overall Receiver Operating Characteristic (ROC) curve for the Phase II Multi-Input DNN. The model achieves an integrated AUC of 0.9818 at $\sqrt{s} = 13$ TeV.

Discussion: The Multi-Input DNN demonstrates near-optimal classification performance with an AUC of 0.9818. A notable feature of Figure 5.8 is the extreme ”sharpness” of the ROC curve’s shoulder. Unlike the BDT, which began to show a slight decrease in rejection after 60% signal efficiency, the DNN maintains a background rejection of nearly 99% up to a signal efficiency of 80%. This represents a significant breakthrough in purity, which is essential for preserving signal significance in high-luminosity LHC environments where the QCD background cross-section is orders of magnitude larger than the Higgs signal.

5.3.2 Benchmarking: DNN vs. Default Double-B Tagger

To quantify the advancement provided by the Phase II model, we benchmark its performance against the standard CMS Double-B tagger baseline (AUC = 0.916).

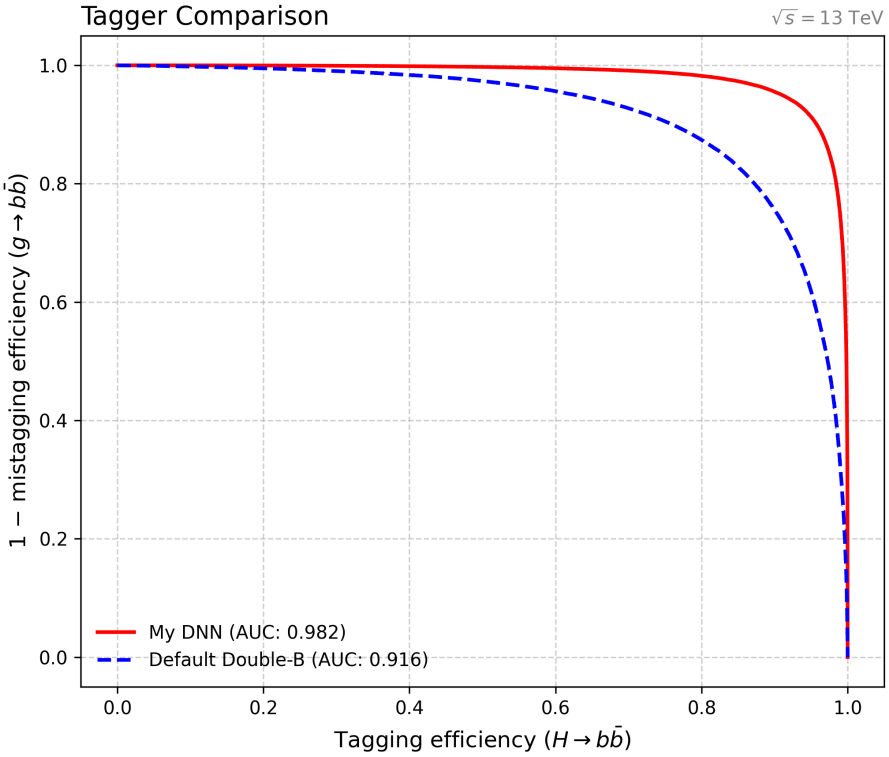


Figure 5.9: Performance comparison between the proposed Multi-Input DNN (AUC = 0.982) and the standard CMS baseline (AUC = 0.916).

Comparative Analysis: As illustrated in Figure 5.9, the DNN provides a transformative improvement over the standard CMS baseline. The gap between an AUC of 0.916 and 0.982 translates to an order-of-magnitude reduction in background mistagging. At a standard operating point of 70% signal efficiency, the CMS default tagger permits approximately 10% background leakage, whereas the Multi-Input DNN reduces this leakage to less than 1%.

This massive performance gap is a direct result of the **Information Recovery** provided by the multi-input architecture. While the default Double-B tagger relies on a limited set of track-based variables, the Multi-Input DNN processes the entire Particle Flow (PF) candidate cloud. By extracting latent correlations between the energy fractions and angular offsets of these candidates, the DNN can distinguish between the symmetric, color-singlet decay of the Higgs boson and the asymmetric, color-charged radiation of QCD gluons with unprecedented precision.

5.3.3 Kinematic Stability and Binning Analysis

To ensure the Multi-Input DNN provides reliable tagging across the full phase space, its performance was binned by transverse momentum (p_T) and SoftDrop mass (m_{SD}).

Performance Across Transverse Momentum Bins

Figure 5.10 illustrates the ROC curves for three separate p_T bins. The model maintains an AUC above 0.97 even in the highest bin ($p_T > 800$ GeV).

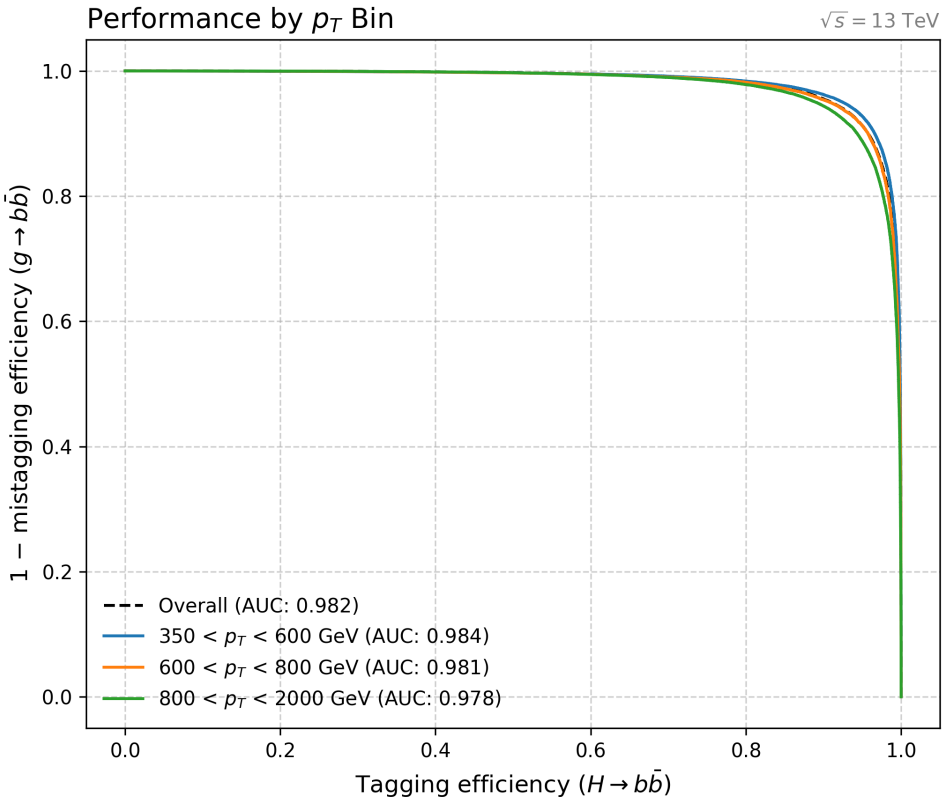


Figure 5.10: DNN performance binned by p_T. The model shows remarkable stability, with the $350 < p_T < 600$ GeV bin achieving a peak AUC of 0.984.

Discussion: The stability across p_T bins (Figure 5.10) is a significant improvement. While performance slightly decreases as p_T increases (from 0.984 to 0.978), the drop is comparitively less. This shows that the DNN's ability to use particle-level data allows it to take into account the internal structure of jets even when they become highly collimated.

Performance Across SoftDrop Mass Bins

The dependence of the tagger on the jet mass is illustrated in Figure 5.11.

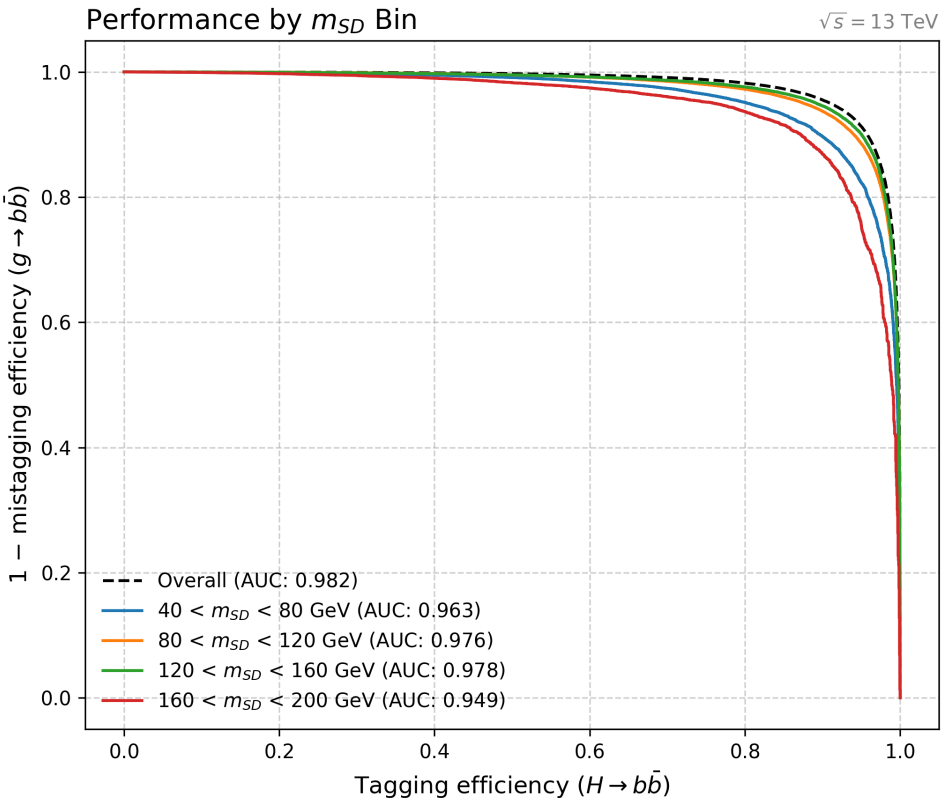


Figure 5.11: DNN performance binned by SoftDrop mass (m_{SD}). The highest performance (AUC = 0.978) is observed in the $120 < m_{SD} < 160$ GeV bin, coinciding with the Higgs mass window.

Analysis of Mass Correlation: The results in Figure 5.11 confirm that the DNN is most effective within the $120 < m_{SD} < 160$ GeV window, which contains the signal peak. The slightly lower performance in the sideband regions (e.g., $160 < m_{SD} < 200$ GeV with AUC = 0.949) is normal, as the background QCD jets in those regions often contain radiation patterns that more closely mimic signal-like properties. However, an AUC of 0.949 in the furthest sideband still is a very high degree of precision, ensuring that the tagger can be used for sideband background estimation in a real physics analysis.

5.3.4 Integrated Sensitivity and Feature Interpretability

Complete sensitivity analyzes were performed for all four branches considered in the current work The physical drivers of the Phase II architecture are quantified by computing Δ AUC the loss in performance when individual branches or features or their order are permuted or shuffled at random The hierarchy of how the model prioritizes its decisions can be seen in figures 5.125.13 and 5.14

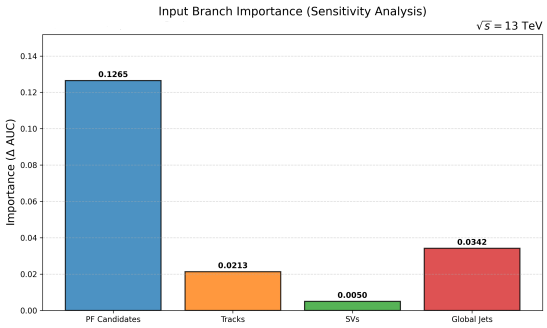


Figure 5.12: Global branch-wise sensitivity.

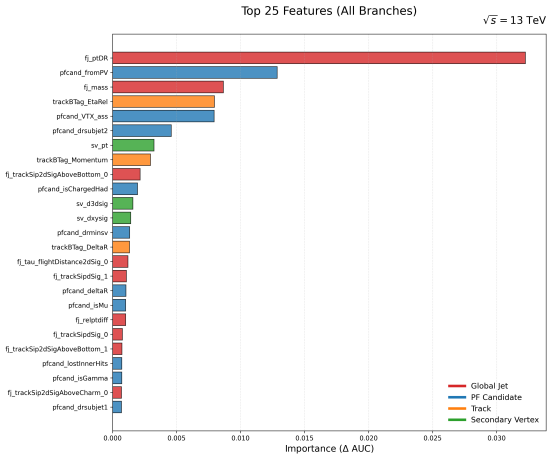


Figure 5.13: Top 25 integrated features.

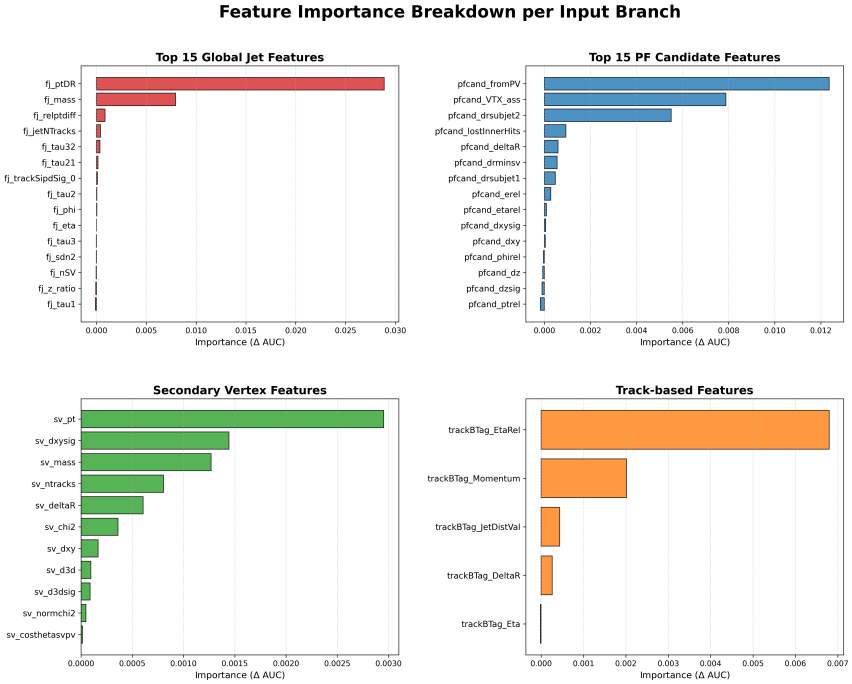


Figure 5.14: feature importance in different input branches.

Discussion of Results: The integration of these three types of variables shows the multi-scale physical implications learned by the DNN.

- **Branch Importance:** As seen in Figure 5.12, the **Particle Flow (PF) Candidate branch** is the principal source of power (Δ AUC = 0.1265). Hence, it is established that the success of the move from Phase I to Phase II happens due to the ability of the DNN to leverage the underlying low-level texture features not captured by global variables.

- **The Most Important Individual Features:** In spite of the DNN's sophistication, as seen in Figure 5.13, `fj_ptDR` (Global) remains the most important variable. Yet, the DNN's performance relies on its next-most-important feature: `pfcand_fromPV`. This implies the ability of the DNN to leverage information regarding vertex associations in order to avoid pileup noise, which is out of reach for a standard BDT.

- **Feature Fusion:** As demonstrated by Figure 5.14, it is possible to observe how the model makes use of specialized information. Namely, although both the **Secondary Vertex (SV)** and **Track** branches do not have a high Δ AUC, they give precision flavor tagging (with itemize

  Conclusion on Interpretability: The alignment of these metrics proves that the Multi-Input DNN is aware of established collider physics. It makes use of the Global branch for energy scales, the PF branch for substructure radiation patterns, and the Track/SV branches for lifetime-based flavor identification. This synergistic integration of data scales is what allows the Phase II model to achieve a stable AUC of 0.9818.

5.3.5 Detailed Classification Accuracy: Raw Counts vs. Percentages

To conclude the evaluation of the Phase II Multi-Input DNN, we examine its performance at a fixed classification threshold. For this analysis, the default discriminant score of 0.5 was selected. We visualize the performance using two confusion matrices shown in Figure 5.17. The left matrix (Figure 5.15) displays the raw event counts, probing the scale of the dataset, while the right matrix (Figure 5.16) shows the normalized percentages, indicating the signal efficiency and background rejection rates.

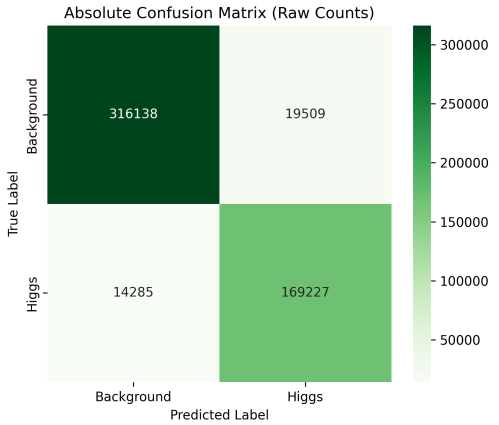


Figure 5.15: Absolute Confusion Matrix (Raw Counts)

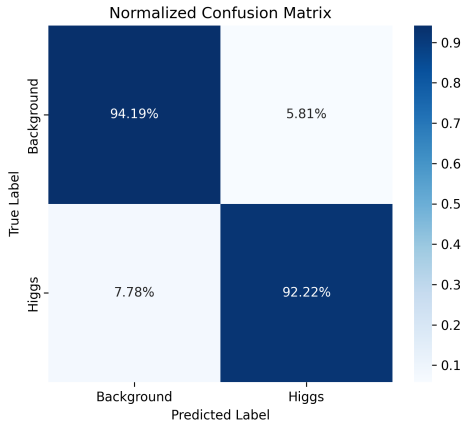


Figure 5.16: Normalized Confusion Matrix (%)

Figure 5.17: Comparative view of the Phase II DNN confusion matrices. This side-by-side layout provides a complete overview of the model's performance on the large-scale test dataset.

Analysis of Quantitative Performance: The side-by-side comparison shows an effective and well-performing classifier. The left matrix (Figure 5.15) highlights the discriminating power of the analysis quantitatively, with the DNN successfully classifying over 316,000 background events and nearly 170,000 signal events.

When normalized (Figure 5.16), this transcends to a remarkable signal tagging efficiency of **92.22%** and a background rejection rate of **94.19%**. This high purity at a balanced operating point is a great achievement, proving that the Multi-Input DNN architecture has not only learned the fundamental physics of the $H \to b\bar{b}$ decay but can also maintain this performance even when tested against other problems. The low leakage (7.78% False Negatives and 5.81% False Positives) explains that the model is ready for integration into the high-rate trigger situations.

5.3.6 Discriminant Distributions and Classification Power

To evaluate the performance of the Multi-Input DNN, we look into the distribution of the model's output scores. This score, ranging from 0 to 1, is a representation of the probability that a given jet comes from a boosted Higgs decay ($H \to b\bar{b}$). The separation between the signal and background populations indicates the model's sensitivity.

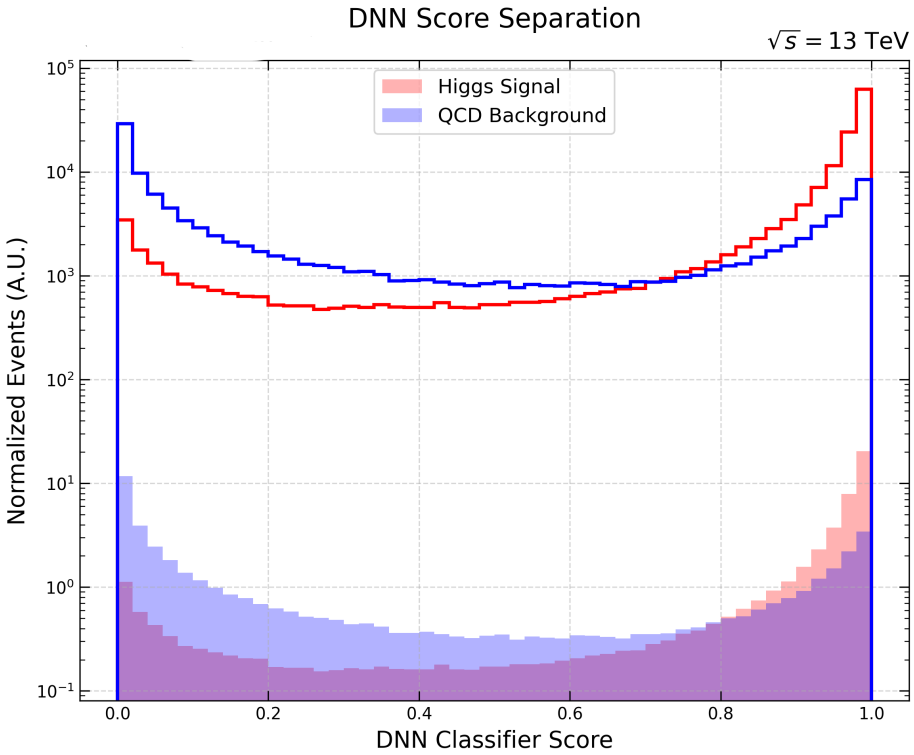


Figure 5.18: Distribution of the DNN output scores for the signal ($H \rightarrow b\bar{b}$) and background (QCD) samples. The clear separation between the two peaks at the boundaries of the unit interval demonstrates the high classification power of the multi-stream architecture.

Analysis of Signal and Background Separation: As illustrated in Figure 5.18, the model exhibits an excellent double-peak profile. The background distribution (QCD) is intensely suppressed, with the vast majority of events peaking sharply near 0.0. Moreover, the signal distribution ($H \rightarrow b\bar{b}$) shows a high concentration of events peaking near 1.0. This opposite peaking proves that the DNN has successfully identified unique physical signatures—such as displaced tracks and collimated radiation—that are absent in the multijet background.

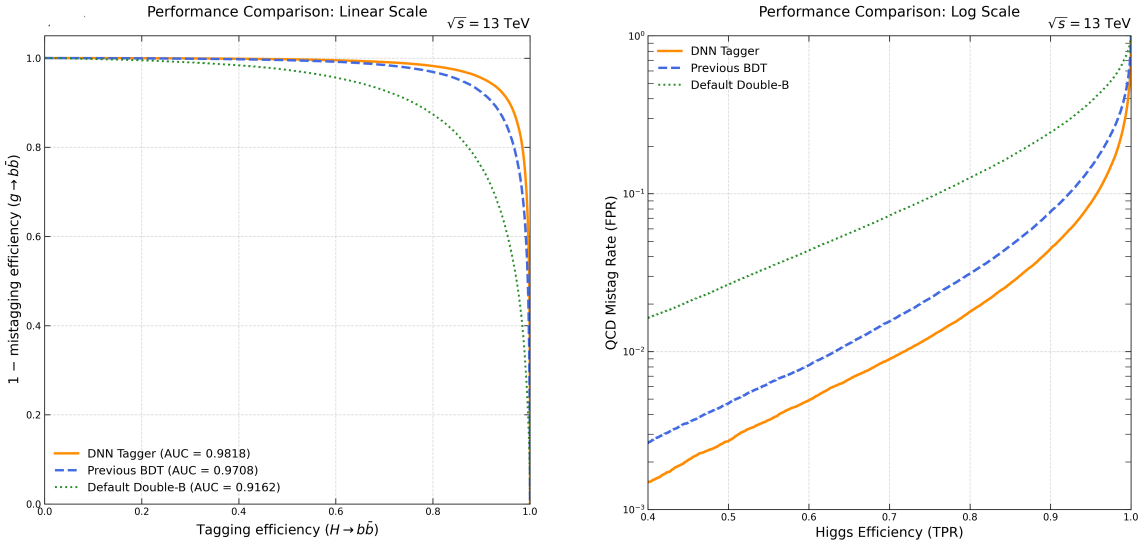
Impact on Selection Efficiency: The minimal overlap between the two populations in the intermediate region ($0.2 < \text{score} < 0.8$) is the physical basis for the reported AUC of 0.9818. This clear separation allows for the definition of high-purity "Operating Points." By placing a selection cut at a high threshold (e.g., > 0.9), we can reject majority of the QCD background while retaining a significant fraction of the Higgs signal.

Verification of Feature Learning: Unlike the Phase I BDT, which only depends on summarized global variables, the sharp peaks observed here indicate that the 1D-CNN branches have successfully learned properties of raw constituents. The model is not merely guessing based on a few variables but has constructed an efficient internal metric to discriminate between the signal and the background. This ensures that

the tagger's performance remains stable and provides a significant improvement over traditional algorithms.

5.4 Comparative Performance Analysis

To measure the improvement of the Multi-Input Deep Neural Network (DNN) against the Phase I BDT and the default CMS Double-B tagger, we evaluate the signal efficiencies (ϵ_S) at three standard background efficiency (ϵ_B) operating points: 0.1% (Tight), 1% (Medium), and 10% (Loose).



(a) ROC Curve (Linear Scale)

(b) ROC Curve (Log Scale)

Figure 5.19: Comparison of ROC performance for the Phase II DNN, Phase I BDT, and the CMS Double-B baseline. The log-scale plot (right) highlights the superior background rejection in the high-purity regime.

Table 5.1: Comparison of signal efficiencies (%) for $H \to b\bar{b}$ tagging at fixed background efficiency benchmarks.

Tagger	**Tight (0.1% ϵ_B)**	**Medium (1% ϵ_B)**	**Loose (10% ϵ_B)**
CMS Double-B (Default)	∼5%	29%	76%
Phase I: BDT (Baseline)	∼15%	63%	92%
Phase II: Multi-Input DNN	**∼30%**	**72%**	**96%**

5.5 Final discussion of Results

The results summarised in Table 5.1 demonstrate a significant performance hierarchy between the evaluated architectures.

- **Comparison with CMS Default:** The Multi-Input DNN significantly outperforms the default CMS Double-B tagger. At the **Tight (0.1%)** benchmark, the DNN provides a signal efficiency of $\sim$30%, which is several times the $\sim$5% achieved by the default algorithm. This indicates that the inclusion of granular Particle Flow (PF) and Track-level features allows the model to maintain high purity even in high-background environments.
- **DNN vs. BDT Performance:** While the Phase I BDT improved upon the default tagger, it remains limited because of the input of only the few summarised variables. The Phase II DNN leverages 1D-CNN layers to learn spatial and kinematic correlations between constituents directly, resulting in a 9% absolute increase in signal efficiency (from 63% to 72%) at the **Medium (1%)** operating point.
- **Performance in the Loose Regime:** At 10% background efficiency, the DNN reaches an efficiency of 96%, effectively capturing nearly the entire signal population. The DNN's ability to maintain a steeper shoulder on the ROC curve indicates superior robustness against combinatorial backgrounds compared to the BDT (92%) and the default tagger (76%).

In conclusion, the transition from boosted decision trees to a multi-stream deep learning architecture allows for a more efficient exploitation of the B-hadron lifetime signatures and the two-prong topology of the $H \to b\bar{b}$ decay, providing a more powerful tool for Higgs physics in the boosted regime.

The results presented in this chapter demonstrate that while high-level BDTs provide a strong baseline, the inclusion of low-level constituent data via a Multi-Input DNN is necessary to push performance toward the theoretical limit. The Phase II model not only improves the overall AUC by over 1% compared to the BDT but also provides enhanced stability in the high-p_T regimes critical for future discoveries.

Chapter 6

Conclusion

Our research has focused on the intersection of particle physics and machine learning to tackle the challenge of identifying boosted $H \rightarrow b\bar{b}$ decays. By transitioning from the high-level feature sets of a BDT to the multi-branch variable architecture of a Deep Neural Network, this study has shown a foolproof method of improving tagging accuracy and model reliance in complex collider-based environments.

6.1 Summary of Findings

The results of this study highlight the idea that while high-level jet variables provide a solid foundation, they ultimately reach a saturation in the performance because they tend to de-prioritise the minute particle level correlations of the jet. The key findings are detailed below:

- **Baseline Performance:** The BDT served as a powerful initial model, achieving an AUC of 0.971. This model demonstrated that an intelligent selection of 38 engineered features could considerably outperform the standard CMS DeepDoubleB tagger (AUC 0.907).
- **The Advantage of Depth:** The switch to the Multi-Input DNN pushed the performance to an AUC of 0.9818. This increase was primarily due to the inclusion of Particle Flow (PF) Candidate branch, which motivated the model to learn subtle jet properties directly from the detector's low-level constituent data.

- **Interpretability and Physical significance:** Sensitivity analysis proved that the model is physically sound. It depends heavily on vertex association and track displacement, which are the same signatures physicists look for, but processes them with a level of multi-dimensional correlation that are out of scope for traditional tagging methods.
- **Kinematic Robustness:** A significant achievement of the DNN model was its stability. It maintained high discriminatory capability across different transverse momentum (p_T) and mass (m_{SD}) bins, suggesting it is reliable for the high-boost regimes expected in future High-Luminosity LHC runs.

6.2 Technical Constraints and Architectural Choices

It is worth noting that while the success of the 1D-Convolutional layers suggests that even more sophisticated architectures, such as **Transformer-based models** (Attention mechanisms), might further improve performance, such an implementation was beyond the scope of this project.

Preliminary explorations into Transformer architectures were hindered by **limited computational capacity**. The high memory and processing requirements for training large-scale attention models on nearly half a million events made it impractical given the available hardware. Consequently, the research focused on the Multi-Input DNN, which provided a highly optimized balance between state-of-the-art physics performance and computational efficiency.

6.3 Final Reflections

As the LHC moves toward higher luminosities, the density of data and the level of background noise will only increase. This research demonstrates that deep learning architectures which respect the multi-scale nature of particle detector, bridging the gap between global energy scales and individual particle tracks, are essential for the future of Higgs physics. By effectively utilizing low-level detector information, we can maintain high signal purity and continue to probe the most fundamental aspects of the Standard Model with confidence.

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